MH4110: Partial Differential Equations

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Preface and Disclaimer

This note is written by me, a student who taken the course MH4110: Partial Differential Equations under Prof. Tong Ping during academic year 25/26 at the School of Physical and Mathematical Sciences of Nanyang Technological University in Singapore.

I hereby disclaim that the contents of this note are intended solely for my personal use and are not originally produced by me, except for the presentation of the definitions and the written proofs. I do not take responsibility for any grammatical or mathematical errors in this note. The note may be incomplete, and once the course is finished, I will no longer update it for any reason.

Chapter 1

Where PDEs Come From

1.1 Notations, Definitions, and Basic Concepts

Throughout the notes we will adapt the following notations and symbol:

- $u_x = \dfrac{\partial u}{\partial x} = \partial_x u$
- $u_{xy} = \dfrac{\partial^2 u}{\partial x \partial y} = \partial_{xy} u$
- $\nabla u = \operatorname{grad} u = u_x \mathbf{i} + u_y \mathbf{j} = (u_x, u_y)$
- $\Delta u = u_{xx} + u_{yy}$
- $\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = u_x + v_y + w_z$ where $\mathbf{F} = (u, v, w)$
- $\nabla \times \mathbf{F} = \operatorname{curl} F = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ u & v & w \end{vmatrix}$
- Given $\vec{v} = (a, b) \neq \mathbf{0}$. The directional derivative of u along $\vec{v}$ at (x, y) is defined by

$$\nabla_{\vec{v}}\, u = \nabla u \cdot \frac{\vec{v}}{|\vec{v}|}$$

We assume that $u_{xy} = u_{yx}$. Some tools that will be useful are listed here:

- Multivariable chain rule
- Derivatives of integrals
- Green's Theorem
- Divergence Theorem

Definition 1.1.1 (Partial Differential Equation)**.** A partial differential equation (PDE) is an identity that relates

- the independent variables (i.e. the inputs): $x, y, \ldots$
- the dependent variable u
- the partial derrivatives of u

A PDE can be expressed in the form of $F(x, y, \ldots, u, u_x, u_y, \ldots, u_{xx}, u_{xy}, u_{yy}, \ldots) = 0$.

Definition 1.1.2 (Order)**.** The order of a PDE is the order of the highest partial derivative that appears in the equation.

Definition 1.1.3 (Solution)**.** A solution of a PDE is a function $u(x, y, \dots)$ that satisfies the equation identically, possibly in some region of $x, y, \dots$ variables.

Definition 1.1.4 (Linear operator)**.** Let $\mathcal{L}$ be an operator. We say $\mathcal{L}$ is a linear operator if the following is satisfied

$$\mathcal{L}(u + v) = \mathcal{L}u + \mathcal{L}v \quad \text{and} \quad \mathcal{L}(cu) = c\mathcal{L}u$$

where u, v are functions and c is constant.

Definition 1.1.5 (Linearity and Homogeneity)**.** Let $\mathcal{L}$ be a linear operator. Then the equation $\mathcal{L}u = g$ is said to be linear, where g is a given function of independent variables. Additionally, we say the equation is homogeneous linear equation if $g = 0$, and vice versa.

Theorem 1.1.6 (Superposition principle)**.** *Consider the equation $\mathcal{L}u = 0$. Suppose that $u_1, u_2, \dots, u_n$ are solutions of the equations. Then the linear combination of the solutions is also a solution:*

$$\sum_{j=1}^{n} c_j u_j(x)$$

where c_j are constants. Additionally, the sum of a homogeneous solution and an inhomogeneous solution is an inhomogeneous solution.

The upshot of superposition principle is that the solution structure of a PDE is always in the form of **general solution + particular solution**, where the general solution is obtained simply solving the corresponding homogenous PDE. On the other hand, the particular solution might requires some hardwork to figure out. If the PDE is homogeneous, then the particular solution degenerates into 0.

1.2 First Order Linear Equation

Here, we will investigate the first-order linear PDE, which takes the form

$$a(x, y)u_x + b(x, y)u_y + c(x, y)u = f(x, y)$$

where a, b, f are continuous functions in some domain D. In particular, the two methods to solve such PDE are geometric methods and coordinate methods.

We will be using directional derivative. To interpret it, suppose given the directional derivative of u along $\vec{v}$ is $\nabla_{\vec{v}} u = g(x, y)$. This means that in the direction of $\vec{v}$, the gradient of u is $g(x, y)$.

<u>Type 1.1: $au_x + bu_y = 0$ with $a^2 + b^2 \neq 0$</u>

We first introduce the geometric method. Firstly, the equation can be rewritten as follow:

$$\begin{aligned} au_x + bu_y &= 0 \\ (a, b) \cdot \nabla u &= 0 \\ \nabla_{(a,b)} u &= 0 \end{aligned}$$

This suggests that $u(x, y)$ is constant along the direction (a, b). Take note that there are three variables here: x, y, and u, so one can picture that in a 3-dimensional cartesian coordinate system.

The line that parallel to (a, b) has the equation $bx - ay = c$, where c is constant. The family of such lines are called the *characteristic lines*. Since u is constant on these lines, the solution is then

$$u(x, y)|_{bx-ay=c} = f(c)$$

This means that if we only examine (x, y) that lies on the line $bx - ay = c$, the output u is an arbitrary function of one variable f. Of course, we can substitute c so that the solution becomes

$$u(x, y) = f(bx - ay)$$

The next method is coordinate method. This method heavily relies on the characteristic lines (or curves), which in this case we have $bx - ay = c$. The strategy is summarised here:

1. Changing variables, where one of the new variable has to be the form of characteristic lines (or curve).
2. Replace all x and y derivatives into the newly defined variables.
3. Rewrite the PDE and solve it.

We use the same example to demonstrate: consider $x' = x$ and $y' = bx - ay$. By chain rule we see

$$\frac{\partial u}{\partial x} = u_{x'}\frac{\partial x'}{\partial x} + u_{y'}\frac{\partial y'}{\partial x} = u_{x'} + bu_{y'}$$
$$\frac{\partial u}{\partial y} = u_{x'}\frac{\partial x'}{\partial y} + u_{y'}\frac{\partial y'}{\partial y} = -au_{y'}$$

Then, rewrite the PDE $au_x + bu_y = 0$ into

$$a(u_{x'} + bu_{y'}) + b(-au_{y'}) = 0$$

which reduces to $au_{x'} = 0$. If $a \neq 0$, then $u_{x'} = 0$, and thus

$$u = f(y') = f(bx - ay)$$

If $a = 0$, then we must have $b \neq 0$. The PDE then becomes $bu_y = 0$, which reduces to $u_y = 0$. This gives $u = f(x)$, which is a specific form of $f(bx - ay)$. To conclude, we have the solution

$$u = f(bx - ay)$$

where f is an arbitrary single variable function.

<u>Type 1.2: $au_x + bu_y = c$ with a, b, c as constants and $a \neq 0$</u>

Recall that from Superposition Principle the solution structure of any PDE is always

$$\text{general solution} + \text{particular solution}$$

Here general solution is already obtained previously. It remains to make an educated guess on form of particular solution. With experience from ODE, it is clear that we can consider $u_0(x, y) = Ax$ where A is some constant. Substituting into the PDE we get

$$Aa + 0 = c \implies A = \frac{c}{a}$$

Thus a particular solution is $u_0 = \frac{c}{a}x$, and the full solution is

$$u = f(bx - ay) + \frac{c}{a}x$$

<u>Type 1.3: $au_x + bu_y = cu$ with a, b, c as constants and $a \neq 0$</u>

Using the characteristic equation method we listed down

$$\frac{dx}{a} = \frac{dy}{b} = \frac{du}{cu}$$

Solving $\frac{dx}{a} = \frac{dy}{b}$ gives us

$$bx - ay = C$$

where C is a constant.

Next, relate dx (or dy) with du. Thus by solving $\frac{dx}{a} = \frac{du}{cu}$ we obtain solution

$$\ln|u| = \frac{c}{a}x + f(C')$$

which is equivalent to $u = e^{\frac{c}{a}x}g(C'')$ by removing the logarithm. Lastly, since C'' is an arbitrary constant, and previously we obtained that $bx - ay = C$ where C is constant. We can substitute C'' so that we now get

$$u = e^{\frac{c}{a}x}g(bx - ay)$$

Remark 1.2.1. Alternatively, one can choose the following method:

1. When $u = 0$, check that it is a solution.
2. When $u \neq 0$, we can divide u to obtain
$$a\frac{u_x}{u} + b\frac{u_y}{u} = c$$
3. By setting $v = \ln|u|$, we see that $v_x = \frac{u_x}{u} + \frac{u_y}{u}$
4. The PDE can be rewritten as $av_x + bv_y = c$, which the solution is fully known.

<u>Type 2: $a(x,y)u_x + b(x,y)u_y = 0$ with $a^2 + b^2 \neq 0$</u>

In the case that the coefficient of u_x and u_y are variables, similar to before, we can solve it in geometric method or coefficient method. The characteristic equation of this type is

$$\frac{dx}{a(x,y)} = \frac{dy}{b(x,y)} \iff \frac{dy}{dx} = \frac{b(x,y)}{a(x,y)}$$

In general the PDE can be solved as long as the ODE above can be solved.

Example 1.2.2. We solve $u_x + yu_y = 0$ in two methods. Regardless of which method we use, we need to solve the characteristic curves:

$$\frac{dy}{dx} = \frac{y}{1} \implies y = Ce^x \implies C = ye^{-x}$$

For the geometric method, we know that $u(x,y)$ is constant on $y = Ce^x$ where C is a constant. Thus $u(x,y) = f(C)$ where f is an arbitrary function of single variable. Rearranging and substituting we get general solution

$$u(x,y) = f(e^{-x}y)$$

For the coefficient method, we let $x' = x$ and $y' = ye^{-x}$. Using chain rule we obtain

$$u_x = u_{x'} - ye^{-x}u_{y'} \quad \text{and} \quad u_y = e^{-x}u_{y'}$$

Next, rewritting $u_x + yu_y = 0$ we get $u'_x = 0$, giving us the solution

$$u = f(y') = f(ye^{-x})$$

Example 1.2.3. We solve $u_x + 2xy^2u_y = 0$. First, the characteristic curve:

$$\frac{dy}{dx} = 2xy^2 \implies x^2 + \frac{1}{y} = C$$

where C is a constant. Note that the solving process of the characteristic curve involves dividing y, thus we need to check separately that whether $y = 0$ is a solution, which it indeed is.

Thus, using the geometric method, we see that the solution of the PDE is

$$u(x,y) = f(C) = f\left(x^2 + \frac{1}{y}\right)$$

Notice that this expression does not degenerate to another solution $y = 0$, thus the complete solution is

$$u(x,y) = f(C) = f\left(x^2 + \frac{1}{y}\right) \quad \text{or} \quad y = 0$$

In summary, first order linear PDE can be fully solved using the method of characteristic, and combined with geometric method or coefficient method. Say given $P(x,y)u_x + Q(x,y)u_y = R(x,y,u)$, the characteristic equations are

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{du}{R}$$

1.3 Flows, Vibrations, and Diffusions

In this part, we introduce some PDEs that describe physical phenomenon. We omit the derivation of most PDEs here. One who is interested can check any standard PDE materials on his own.

Specifically, the three fundamental PDEs of this course are:

- Wave equation, which describes vibrations.
- Heat equation, which describes flows.
- Laplace equation, which describes diffusions.

Transport Equation: $u_t(x,t) + cu_x(x,t) = 0$

Transport equation describe the density of cars on a road way under ideal conditions. We will derive this equation here.

We assume that the road is long and straight, with all the cars running at the same speed c. Also we assume that no cars entering or exiting the road. Define $u(x,t)$ be the car density at time t and position x.

The number of cars in the interval $[0,b]$ at time t can be calculated as

$$M = \int_0^b u(x,t)dx$$

Since all the cars are in constant speed c, at the later time $t+h$ they move to the right by ch km. Thus we have

$$M = \int_0^b u(x,t)dx = \int_{ch}^{b+ch} u(x,t+h)dx$$

where differentiating both sides wrt b we get

$$u(b,t) = u(b+ch, t+h)$$

By changing the specific location b to an arbitrary position x and rearranging the equation we get

$$u(x+ch, t+h) - u(x,t) = 0$$

We are interested at the car density at some sudden time instead of a time interval. So we divide h and take limit $h \to 0$:

$$\begin{aligned}
\lim_{h\to 0} \frac{u(x+ch,t+h) - u(x,t)}{h} &= 0 \\
\lim_{h\to 0} \frac{u(x+ch,t+h) - u(x+ch,t) + u(x+ch,t) - u(x,t)}{h} &= 0 \\
\lim_{h\to 0} \frac{u(x+ch,t+h) - u(x+ch,t)}{h} + \lim_{h\to 0} \frac{u(x+ch,t) - u(x,t)}{h} &= 0 \\
\frac{\partial u(x,t)}{\partial t} + c\frac{\partial u(x,t)}{\partial x} &= 0
\end{aligned}$$

This gives us

$$u_t(x,t) + cu_x(x,t) = 0$$

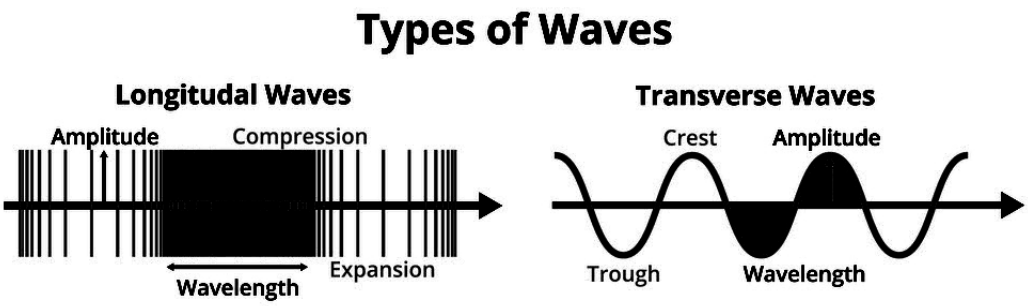
This is a first order linear PDE. The characteristic line is $ct - x = W$ where W are arbitrary constants, and the general solution is $f(ct-x)$ where f is arbitrary function.

Remark 1.3.1. Of course we can check if the derived equation is consistent or not, in the sense that if the units of two terms agree. Here u_t has the unit of numbers of car per second. Recall c is speed, so it has unit meter per second. Also u_x has unit of numbers of car per meter squared (due to the extra derivative $\partial/\partial x$). Together cu_x has unit of numbers of car per second.

Remark 1.3.2. In the case that the car moves to left, then c is taken to be negative.

Wave Equation: $u_{tt} = c^2 u_{xx}$

This is first of the three fundamental PDEs of this course. Wave equation gives the PDE that describe, clearly, waves. In general, there are two types of waves in 2D: transverse wave and longitudinal wave.



The wave equation introduced here describes transverse wave and assume that no involmen of longitudinal wave. First, we introduce several variables:

- Linear density ρ: unit of mass per unit of length.
- Then ρdx is then the unit of mass of length segment dx
- Let $u(t, x)$ be the displacement of strings from equilibrium position at time t and position x.
- We assume the linear density is constant throughout the string.
- We ignore all other forces except for the tension $\mathbf{T}(x, t)$.

Definition 1.3.3 (Wave Equation)**.** Assuming that T is a constant, the wave equation takes the form

$$u_{tt} = c^2 u_{xx}$$

where $c = \sqrt{\frac{T}{\rho}}$ is the wave speed.

Remark 1.3.4. We listed some variants of wave equations here.

- If considering air resistance r, the wave equation is then

$$u_{tt} - c^2 u_{xx} + r u_t = 0$$

- If there is a traverse elastic force, the wave equation becomes

$$u_{tt} - c^2 u_{xx} + k u = 0$$

- If there is an external force f, the wave equation becomes

$$u_{tt} - c^2 u_{xx} = f(x, t)$$

- There is wave equation for multidimensional case, but will not be mentioned here.

Heat Equation: $u_t = k(u_{xx} + u_{yy} + u_{zz})$

This is the second of the three fundamental PDEs of this course. Despite suggested by its name, heat equation can describe general flows. Again, we first introduce the variables needed:

- We work in a Minskowski space, i.e. spatial dimension (x, y, z) that describes the position and dimension t that describe the time. This is equivalent to $\mathbb{R}^4$.
- Let $u(x, y, z, t)$ be the temperature at point (x, y, z) and time t.

- Let D be a domain in $\mathbb{R}^3$ that represents the body, and $H(t)$ be the amount of heat (or energy) in D.
- Let ρ be the density of the material.
- Let c be the specific heat of the material (In physics, specific heat is the energy needed to raise one unit of mass of a substance by one unit of temperature. Simply speaking, it is just a constant.)
- Let κ be the heat conductivity of the material. (Another constant. Simply speaking, it descirbes the material's ability to conduct heat. Larger values represents better conductivity.)

For the sake of simplicity, we assume that the space ($\mathbb{R}^3$) outside the body (D) is colder. With that said, the direction of the heat flow, or more accurately, the energy flux, go from inside to outside.

Definition 1.3.5 (Heat Equation)**.** With all defined notations, the heat equation takes the form

$$c\rho u_t = \nabla \cdot (\kappa \nabla u)$$

In the case that c, ρ and κ are constant, we can combine the constant into $k = \frac{\kappa}{c\rho}$ and the equation becomes

$$u_t = k\triangle u$$

where recall $\triangle u = u_{xx} + u_{yy} + u_{zz}$.

Ultimately, the heat equation says that the heat flow u_t is propotional to $\triangle u$.

Remark 1.3.6 (Interepretation of heat equation.)**.** In 1D, we learned that the second derivative represents the curvature of the curve: concave or convex. In particular, if $\triangle u = u_{xx}$

- $= 0$, it means that it is a straight line. (The point is the same as its neighbors).
- > 0, it means that concave upwards ($\cup$ shape, the point is lower than its neighbors).
- < 0, it means concave downwords ($\cap$ shape, the point is higher than its neighbors)

In the 3D case, the expression $\triangle u = u_{xx}+u_{yy}+u_{zz}$ carries the same interpretation. The only difference is that we are now talking about 'temperature', so the higher the values of $|\triangle u|$, the larger the temperature difference between the point and its surroundings neighbors, and thus it is expected to have higher heat flow, i.e. higher u_t.

<u>Laplace equation: $u_{xx} + u_{yy} + u_{zz} = 0$</u>

Laplace equation describes diffusions. Think of a room with a heat source, it is clear that the heat is not homogeneous in the room: the closer the heat sources, the higher the heat. Assuming that the heat source exists in the room for long enough time, the temperature of the room is eventually homogeneous. In this case, the heat does not change with time, thus the heat flow u_t is 0. Therefore from heat equation we obtain:

Definition 1.3.7 (Laplace equations)**.** With same notation defined previously, the Laplace equation takes the form

$$\triangle u = u_{xx} + u_{yy} + u_{zz} = 0$$

and its solutions are called harmonic solutions.

In the case that the above PDEs is inhomogeneous, i.e. $\triangle u = f$ for some non-zero f, we called it the *Poisson equation.*

1.4 Initial and Boundary Conditions

As in ODE, the solution we get from a PDE actually comprises a whole family of functions that satisfies the given equation. In reality, some conditions are required to single out a solution. These conditions fall into two types: initial conditions and boundary conditions.

Intial condition specify the physical state at a particular time t_0. Simply speaking, it gives what happens exactly at same positions and time.

On the other hand, *boundary condition* gives a domain D for which the PDE is defined.

Example 1.4.1 (Examples of boundary conditions.)**.** Let D be a domain where a given PDE is defined. Let $\mathbf{n}$ be the unit normal vector pointing outwards on ∂D. Let $\Gamma \subseteq \partial D$.

- Dirichlet boundary condition: $u(\mathbf{x}, t) \mid_\Gamma = g(\mathbf{x}, t)$
- Neumann boundary condition: $u_{\mathbf{n}} \mid_\Gamma = g(\mathbf{x}, t)$
- Robin boundary condition: $(u_{\mathbf{n}} + au)|_\Gamma = g(\mathbf{x}, t)$

In any of the above example, we say the boundary conditions are homogeneous if $g(\mathbf{x}, t) = 0$, and inhomogeneous otherwise.

1.5 Well-posedness of a PDE

Suppose given a PDE in a domain D with a set of initial and/or boundary conditions. We say it is well-posed if all the following are met:

1. (Existence) There exists at least one solution.
2. (Uniqueness) There exists a unique solution.
3. (Stability) Small changes in the initial and/or boundary conditions lead to small changes in the output.

Well-posedness is an important property to have if one is using PDE to study the actual world. In particular, existence and uniqueness ensures that the model(PDE) is reliable, and stability gives basic control of the system due to continuous dependence.

1.6 Types of Second-Order Equations

We have investigated first-order linear PDE in the previous section, where we demonstrate how it can be reduced to an ODE problem via the method of characteristic curves. We now study second-order linear PDE. We will only go through the types of PDEs in this part, and the solution of each PDE will be presented in the next chapter.

Definition 1.6.1 (Second-order linear PDE)**.** A second-order linear PDE with n variables $x_1, x_2, \ldots, x_n$ is a PDE that takes the form

$$\sum_{i,j=1}^{n} a_{ij} u_{x_i x_j} + \sum_{i=1}^{n} a_i u_{x_i} + a_0 u = 0 \tag{1.1}$$

where all constants a_{ij}, a_i, and a_0 are reals. Recall that we assumed that mixed derivatives are equal, thus we have $a_{ij} = a_{ji}$.

Definition 1.6.2 (Coefficient Matrix and Eigenvalues of PDEs)**.** Suppose given a second-order linear PDE of form (1.1). Its coefficient matrix is defined to be the $n \times n$ matrix where the (i, j)-th entry is the coefficient a_{ij}. The eigenvalues of a PDE is the eigenvalues of its coefficient matrix.

Definition 1.6.3 (Elliptic, Hyperbolic, Ultrahyperbolic, and Parabolic PDEs)**.** Suppose given a second-order linear PDE of form (1.1). We have the following classification of a given PDE:

- Elliptic: if all its eigenvalues are positive or negative.
- Hyperbolic: if all its eigenvalues are non-zero, and one of them has opposite sign from the other $n - 1$ eigenvalues.
- Ultrahyperbolic: if all its eigenvalues are non-zero, and at least two of them are positive as well as at least two of them are negative.
- Parabolic: if exactly one of them is zero, and all others have the same sign.

Note that wave equation, heat equation, and Laplace equation are all second-order linear PDEs. We can now have a classification of them

PDEs	Form	Eigenvalues	Types
Wave equation	$u_{tt} - c^2(u_{xx} + u_{yy} + u_{zz}) = 0$	$\left\{1^{(1)}, -c^{2^{(3)}}\right\}$	Hyperbolic
Heat equation	$u_t - k(u_{xx} + u_{yy} + u_{zz}) = 0$	$\left\{0^{(1)}, -k^{(3)}\right\}$	Parabolic
Laplace equation	$u_{xx} + u_{yy} + u_{zz} = 0$	$\left\{1^{(4)}\right\}$	Elliptic

From linear algebra we know that the determinant $\det(A)$ of a matrix A equals to the product of all its eigenvalues. The following theorem is obtained from this relation:

Theorem 1.6.4. *Given a second-order linear PDE with two variables:*

$$a_{11}u_{xx} + 2a_{12}u_{xy} + a_{22}u_{yy} + a_1u_x + a_2u_y + a_0u = 0$$

Denote its coefficient matrix as A. We have:

1. *If* $\det(A) > 0$, *then the PDE is of elliptic type.*
2. *If* $\det(A) < 0$, *then the PDE is of hyperbolic type.*
3. *If* $\det(A) = 0$, *then the PDE is of parabolic type.*

Improtant: This determinant test does not apply for second-order linear PDE with more than two variables!

Chapter 2

Waves and Diffusions

2.1 The Wave Equation

We have defined wave equation is the previous chapter. Note that if we take the wave equation on $\mathbb{R}$, it takes the form of

$$u_{tt} = c^2 u_{xx}$$

where $x \in \mathbb{R}$ is a variable of reals, and the constant $c > 0$ is the wave speed. We will study its solution in this part.

<u>Method 1: Factorization of the differential operator</u>

Note that we can rewrite the equation into

$$u_{tt} - c^2 u_{xx} = \left(\frac{\partial}{\partial t} - c\frac{\partial}{\partial x}\right)\left(\frac{\partial}{\partial t} + c\frac{\partial}{\partial x}\right) u = 0$$

where what we done is to factorize the differential operators.

Next, define

$$\begin{aligned} v &= \left(\frac{\partial}{\partial t} + c\frac{\partial}{\partial x}\right) u \\ &= u_t + cu_x \end{aligned}$$

The wave equation can then be rewritten as

$$\left(\frac{\partial}{\partial t} - c\frac{\partial}{\partial x}\right) v = 0 \implies v_t - cv_x = 0$$

This is a first-order linear PDE, which its solution is fully known.

By the method of characteristic, we see that

$$v(x,t) = h(x+ct)$$

where h is an arbitrary function. Recovering v we see

$$u_t + cu_x = h(x+ct)$$

This is again a first-order linear PDE, except that it is inhomogeneous. To solve this, we have to solve its general solution and particular solution.

Its general solution is $u = g(x-ct)$ where g is an arbitrary function. For the particular solution, we can verify that $u = f(x+ct)$ is a particular solution:

$$h(x+ct) = u_t + cu_x = cf'(x+ct) + cf'(x+ct) = 2cf'(x+ct)$$

where we see we can take function f so that $f'(x+ct) = \frac{1}{2c}h(x+ct)$.

Therefore the solution is

$$u(x,t) = f(x+ct) + g(x-ct)$$

where f and g are arbitrary functions.

Example 2.1.1. Suppose given $u_{xx} - 3u_{xt} - 4u_{tt} = 0$. This can be viewed as the same to factorize

$$x^2 - 3xt - 4t^2 = 0$$

which is $(x-4t)(x+t) = 0$. Translating back, the given PDE can be factorized as

$$(\partial_x - 4\partial_t)(\partial_x + \partial_t)u = 0$$

To continue solving the PDE, simply let $v = (\partial_x + \partial_t)u$ so that the PDE can be rewritten as

$$v_x - 4v_t = 0$$

The remaining process to continue the solution is the same as described in the above paragraph, the details are omitted here.

<u>Method 2: Characteristic Coordinates</u>

We can introduce the characteristic coordinates

$$\xi = x + ct \quad \text{and} \quad \eta = x - ct$$

By chain rule, we obtain

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial \xi} + \frac{\partial}{\partial \eta} \quad \text{and} \quad \frac{\partial}{\partial t} = c\frac{\partial}{\partial \xi} - c\frac{\partial}{\partial \eta}$$

as well as

$$\frac{\partial^2}{\partial x^2} = \frac{\partial^2}{\partial \xi^2} + 2\frac{\partial^2}{\partial \xi \partial \eta} + \frac{\partial^2}{\partial \eta^2} \quad \text{and} \quad \frac{\partial^2}{\partial t^2} = c^2\left(\frac{\partial^2}{\partial \xi^2} - 2\frac{\partial^2}{\partial \xi \partial \eta} + \frac{\partial^2}{\partial \eta^2}\right)$$

Substituting these into wave equation we get

$$u_{tt} - c^2 u_{xx} = -4c^2 u_{\xi\eta} = 0 \implies u_{\xi\eta} = 0$$

Solving out, we get solution

$$u = f(\xi) + g(\eta) = f(x+ct) + g(x-ct)$$

where f and g are two arbitrary functions.

Remark 2.1.2. The above methods actually suggest that the wave equation has two families of characteristic lines: $x \pm ct =$ constant. As the time t increases:

- $g(x-ct)$ is a wave that travels to the right at speed c.
- $f(x+ct)$ is a wave that travels to the left at speed c.

As wave equation is motivated from physical phenomenon, it is natural to discuss how to solve its IVP. Suppose given the initial condition

$$u(x,0) = \phi(x) \quad \text{and} \quad u_t(x,0) = \psi(x) \quad \text{and} \quad -\infty < x < \infty$$

where ϕ and ψ are arbitrary functions of single variable x, we want to give the exact solution to the wave equation. Since the solution $u = f(x+ct) + g(x-ct)$ is known, by substituting specific x and/or t, and taking derivative when necessarily, we obtain:

$$u(x,0) = f(x) + g(x) = \phi(x) \implies f'(x) + g'(x) = \phi'(x)$$

and

$$u_t(x,0) = cf'(x) - cg'(x) = \psi(x)$$

These two equations give

$$\begin{cases} f'(x) &= \frac{1}{2}\phi'(x) + \frac{1}{2c}\psi(x) \\ g'(x) &= \frac{1}{2}\phi'(x) - \frac{1}{2c}\psi(x) \end{cases}$$

Integrating, we get

$$\begin{cases} f(x) &= \frac{1}{2}\phi(x) + \frac{1}{2c}\int_0^x \psi(s)ds + A \\ g(x) &= \frac{1}{2}\phi(x) - \frac{1}{2c}\int_0^x \psi(s)ds + B \end{cases}$$

Taking sum of two equations, we get $f(x) + g(x) = \phi(x) + (A + B)$. However, by comparing with the initial condition $u(x, 0) = f(x) + g(x) = \phi(x)$, we see that $A + B = 0$.

Lastly, substitute the obtained f and g into the solution $u(x, t) = f(x + ct) + g(x - ct)$ and simply into

$$u(x,t) = \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)ds$$

This is the solution formula for the IVP of wave equation. In particular, the setup of the IVP is called *Cauchy Problem*, and the obtained solution formula is called the *d'Alambert Formula*.

Remark 2.1.3 (General d'Alambert Formula)**.** If the given IVP is not starting at $t = 0$, say $t = t_0 > 0$, one can perform a change of variable by setting $t' = t - t_0$, so that the d'Alambert Formula can be applied to solve the IVP. In general, the formula for solving such IVP is

$$u(x,t) = \frac{1}{2}[\phi(x + c(t - t_0)) + \phi(x - c(t - t_0))] + \frac{1}{2c}\int_{x-c(t-t_0)}^{x+c(t-t_0)} \psi(s)ds$$

Again, it is empahsized that $\phi(x-ct)$ is travelling to the right, and $\phi(x+ct)$ is travelling to the left.

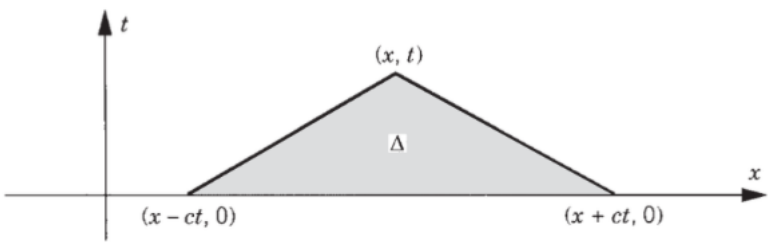
2.2 Causality and Energy

We have discussed d'Alambert formula previously. Note that $u(x, t)$ depends on the values of ϕ and ψ. In particular, the value $u(x, t)$ is affected by the following:

- value of ϕ at point $(x - ct, 0)$
- value of ϕ at point $(x + ct, 0)$
- value of ψ on the interval $[x - ct, x + ct]$.

The interval $[x - ct, x + ct]$ seems important as all informations are contained here.

Definition 2.2.1 (Interval of Dependence, Domain of Dependence)**.** The interval of dependece for point (x, t) is defined to be the interval $[x - ct, x + ct]$. The domain of dependence (or the past history) of the point (x, t) refers to the triangular region with vertices $(x - ct, 0), (x - ct, 0)$, and (x, t)



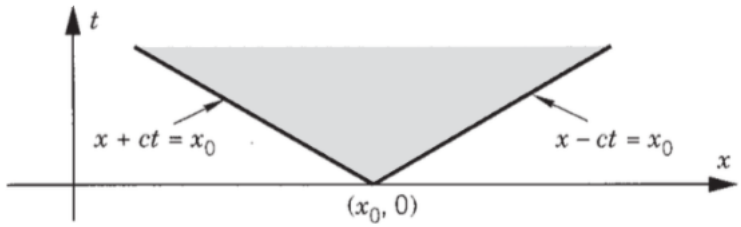
Remark 2.2.2. The triangular region is called the past history of point (x, t) for a reason. Consider that at the moment when the time equals to t, and we are in position x. This makes that everything within the triangular region representing 'the past' from when the time is t, since the second coordinate of all point within the triangular region is smaller than t.

Another physical interpretation of the triangular region is that, only the points inside the triangular region affects the value of $u(x, t)$, and hence the name domain of dependence. If something outside the region is altered, it won't affect the value of $u(x, t)$.

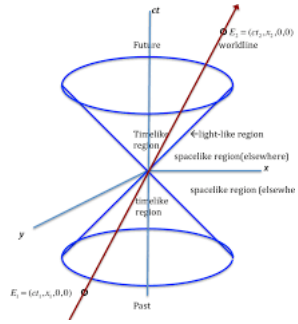
We can change our perspective by asking

> Suppose given a starting point $(x_0, 0)$, for which point (x, t) where the value of $u(x, t)$ will be affected by the given point?

Clearly, many points will be affected, and these points are exactly the points within the shaded region in the following diagram:



Definition 2.2.3 (Region of Influence)**.** The region of influence of the point $(x_0, 0)$ is defined to be the shaded region above.



The value $u(x,t)$ is affected by $(x_0, 0)$ if and only if

$$x - ct \leq x_0 \leq x + ct$$

given that the speed of the wave is bounded by c.

Remark 2.2.4 (Minkowski's light cone)**.** Since light has the property of wave-particle duality, photons obey the wave equation. The Minkowski's light cone, which illustrates causality from past and future, is related by the above concept.

Definition 2.2.5 (Kinetic Energy, Potential Energy, and Total Energy)**.** Consider an infinite string with constant linear density ρ and tension magnitude T. Inheritting the above notation:

- The kinetic energy at time t is defined as

$$E_K(t) = \frac{1}{2}\rho \int_{-\infty}^{\infty} u_t^2(x,t)\ dx$$

- The potential energy at time t is defined as

$$E_P(t) = \frac{1}{2}T \int_{-\infty}^{\infty} u_x^2(x,t)\ dx$$

The total energy at time t is just the sum of the kinetic energy and potential energy, both at time t:

$$E(t) = \frac{1}{2} \int_{-\infty}^{\infty} (\rho u_t^2 + T u_x^2)\ dx$$

Remark 2.2.6 (Conservation of Energy)**.** It is well-known that energy is conserved in a closed system. Mathematically, it means that the rate of change of total energy is 0. Thus, showing the conservation of energy is equivalent to showing that $E'(t) = 0$.

So, we first obtain

$$E'(t) = \int_{-\infty}^{\infty} (\rho u_t u_{tt} + T u_x u_{xt})\ dx$$

Note that the wave equation gives $u_{tt} = c^2 u_{xx}$, where expanding out $c = \sqrt{\frac{T}{\rho}}$ we see that $\rho u_{tt} = T u_{xx}$. Substituting we get

$$E'(t) = \int_{-\infty}^{\infty} (T u_t u_{xx} + T u_x u_{xt})\ dx = T \int_{-\infty}^{\infty} (u_t u_{xx} + u_x u_{xt})\ dx$$

Note that $u_t u_{xx} + u_x u_{xt} = \dfrac{\partial}{\partial x}(u_t u_x)$, again substituting we get

$$E'(t) = T \int_{-\infty}^{\infty} \frac{\partial}{\partial x}(u_t u_x)\ dx$$

By Fundamental Theorem of Calculus, it can be evaluated as

$$E'(t) = Tu_t u_x \Big|_{-\infty}^{\infty}$$

Of course, we assume that the energy is finite, so the wave flatten and rest eventually if we moved far enough away from the source $x = 0$. Recall that u_t represents velocity and u_x represents the slope of the wave. At $x = \pm\infty$, the string is now flat and still, which implies $u_x = 0$ and $u_t = 0$ respectively. Thus

$$E'(t) = Tu_t u_x \Big|_{-\infty}^{\infty} = 0 - 0 = 0$$

The above can be summarized as follow:

Theorem 2.2.7 (Conservation of Energy and IVP)**.** *Adapting the same notation, we see that*

$$E(t) = constant \quad \forall t \geq 0$$

In addition, suppose given IVP $u(x,0) = \phi(x)$ and $u_t(x,0) = \psi(x)$, then

$$E(t) = E(0) = \frac{1}{2}\int_{-\infty}^{\infty} \left(\rho\psi^2(x) + T(\phi'(x))^2\right)\, dx$$

2.3 The Diffusion Equation

The goal here is to solve the diffusion equation:

$$u_t = ku_{xx}$$

and obtain a solution formula, depending on the given initial data $u(x,0) = \phi(x)$. This is more difficult to solve comparing to the wave equation. The strategy is as follow:

1. Show the uniqueness of the solution.
2. Construct a solution to the problem.

We first give a theorem without proof:

Theorem 2.3.1 (Weak Maximum Principle)**.** *If $u(x,t)$ satisfies the diffusion equation $u_t = ku_{xx}$ in a rectangle $D = \{(x,t) : 0 \leq x \leq \ell,\ 0 \leq t \leq T\}$ in space-time, then the maximum value of $u(x,t)$ is achieved either initially at $t = 0$, or on the lateral sides, i.e. $x = 0$ or $x = \ell$.*

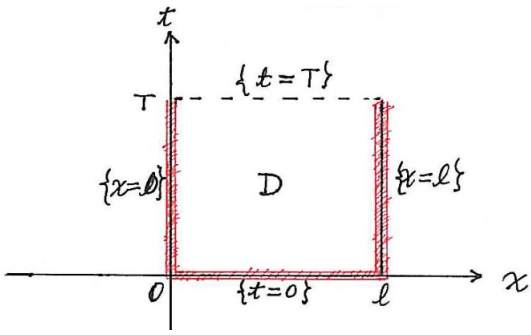
In other words, denote the union of the left, bottom, and right boundaries as:

$$\Gamma := \{(x,0) : 0 \leq x \leq \ell\} \cup \{(0,t) : 0 \leq t \leq T\} \cup \{(\ell,t) : 0 \leq t \leq T\}$$

Then

$$\max_{(x,t)\in\Gamma} u(x,t) = \max_{(x,t)\in D} u(x,t)$$

The notation of Γ is used in the later proof as well. In $x-t$ coordinate system, the union Γ of boundaries looks like this:



Remark 2.3.2.

1. The Strong Maximum Principle says that, the maximum *can and only can* achieved at Γ.
2. Note that Maximum Principle also implies Minimum Principle by applying it to the function $-u(x,t)$ by noting that $(-u)_t = k(-u)_{xx}$ and utilize the identity
$$\min\{u(x,t)\} = -\max\{-u(x,t)\}$$
3. From physical perspective, if applying to heat equation, the principle says that:

   The highest temperature in the interior body cannot exceed the highest initial temperature, or the highest temperature on the boundary.

We now show the uniqueness:

Theorem 2.3.3 (Uniqueness of Dirichlet Problem for Diffusion Equation)**.** *Consider the diffusion equation with Dirichlet boundary conditions:*

$$\begin{cases} u_t - ku_{xx} &= f(x,t) \ ,0 < x < \ell, t > 0 \\ u(x,0) &= \phi(x) \quad ,0 < x < \ell \\ u(0,t) &= g(t) \quad ,u(\ell,t) = h(t) \end{cases}$$

where f, ϕ, g, h are given functions. Then the above equation has unique solution.

Proof using Maximum Principle. Suppose that there are two solutions $u_1(x,t)$ and $u_2(x,t)$. Let $w := u_1 - u_2$, then w satisfies the homogenous equation with zero initial-boundary conditions:

$$\begin{cases} w_t - kw_{xx} &= 0 \quad ,0 < x < \ell, t > 0 \\ w(x,0) &= 0 \quad ,0 < x < \ell \\ w(0,t) &= 0 \quad ,w(\ell,t) = 0 \end{cases} \tag{2.1}$$

For any $T > 0$, define the rectangle $D = \{(x,t) : 0 \le x \le \ell, 0 \le t \le T\}$. Since w satisfeis the diffusion equation in the defined rectangle D, by Maximum Principle we see that

$$w(x,t) \le \max_{(x',t')\in\Gamma} w(x',t') \quad \forall (x,t) \in D$$

Note that for any $(x',t') \in \Gamma$ it takes the form of $(x',0)$, $(0,t')$, or (ℓ,T), which by Equation (2.1) we see that $w(x',t') = 0$ regardless of which form. Together, we see that

$$w(x,t) \le 0 \quad \forall (x,t) \in D$$

Similarly, we can show using Minimum Principle that

$$w(x,t) \ge \min_{(x',t')\in\Gamma} w(x',t') = 0 \quad \forall (x,t) \in D$$

This shows that $u_1(x,t) - u_2(x,t) = w(x,t) = 0$, implying that $u_1(x,t) = u_2(x,t)$ for all $0 < x < \ell$ and all $t > 0$. Thus the solution is unique. □

Proof using Energy Method. Suppose that there are two solutions $u_1(x,t)$ and $u_2(x,t)$. Let $w := u_1 - u_2$, then w satisfies the homogenous equation with zero initial-boundary conditions:

$$\begin{cases} w_t - kw_{xx} &= 0 \quad ,0 < x < \ell, t > 0 \\ w(x,0) &= 0 \quad ,0 < x < \ell \\ w(0,t) &= 0 \quad ,w(\ell,t) = 0 \end{cases} \tag{2.2}$$

The energy of w at time t is defined as

$$E[w](t) = \frac{1}{2}\int_0^\ell [w(x,t)]^2 \, dx$$

which is non-negative. Note that the initial condition $w(x,0) = 0$ implies $E[w](0) = 0$. Differentiating the energy w.r.t. time t, we get

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \int_0^\ell ww_t \, dx$$

Note diffusion equation says $w_t = kw_{xx}$, substituting we get

$$\frac{\mathrm{d}E}{\mathrm{d}t} = k\int_0^\ell ww_{xx} \, dx$$

We can rewrite the integrand by noticing $ww_{xx} = \dfrac{\partial}{\partial x}(ww_x) - w_x^2$, and deriving it to get:

$$\begin{aligned}
\frac{\mathrm{d}E}{\mathrm{d}t} &= k\int_0^\ell ww_{xx} \, dx \\
&= \int_0^\ell k\left(\frac{\partial}{\partial x}(ww_x) - w_x^2\right) \, dx \\
&= k[w(x,t)w_x(x,t)]_{x=0}^{x=\ell} - \int_0^\ell w_x^2 dx \\
&\leq k[w(x,t)w_x(x,t)]_{x=0}^{x=\ell}
\end{aligned}$$

where the last inequality is obtained by noting that $\int_0^\ell w_x^2$ is non-negative. Lastly, since $w(0,t) = 0$ ad $w(\ell,t) = 0$, thus $k[w(x,t)w_x(x,t)]_{x=0}^{x=\ell} = 0$. Together we have

$$\frac{\mathrm{d}E}{\mathrm{d}t} \leq 0$$

This shows that $E[w](t)$ is a decreasing quantity. By the definition of decreasing, for all t we have

$$0 \leq E[w](t) \leq E[w](0) = 0$$

which forces that $E[w](t) = 0$. By definition of E, it implies that $w = 0$ and thus $u_1 = u_2$. Therefore the solution is unique. □

Example 2.3.4. Consider the diffusion equation $u_t = u_x x$ with $(x,t) \in D := \{(x,t) : 0 < x < \ell, t > 0\}$ with boundary conditions:

$$u(0,t) = u(\ell,t) = 0 \quad \text{and} \quad u(x,0) = 4x(\ell - x)$$

We want to show that $0 < u(x,t) < \ell^2$ for all $(x,t) \in D$. By Maximum Principle, the maximum of u appears at the boundary Γ. The given condition $u(0,t) = u(\ell,t) = 0$ implies that the maximum does not occur on the lateral sides, since $u(x,0) = 4x(\ell - x)$ is greater than 0 at some x. It is easy to justify that the maximum of $4x(\ell - x)$ occurs at $x = \ell/2$, where $u(\ell/2, 0) = \ell^2$. Therefore by Maximum Principle we have $u(x,t) < \ell^2$ for all $(x,t) \in D$.

On the other hand, by Minimum Principle, the minimum of $u(x,t)$ must be achieved in Γ. Since $0 \leq x \leq \ell$ in Γ, so $u(x,0) = 4x(\ell - x) > 0$. But $u(0,t) = u(\ell,t) = 0$, thus 0 is the minimum value by Minimum Principle. Together $u(x,t) > 0$ for all $(x,t) \in D$.

Next, we use the energy method to show that $\int_0^\ell u^2 \, dx$ is a strictly decreasing function of t. Define

$$w(t) := \int_0^\ell u^2(x,t) \, dx$$

It is equivalent to show that $\dfrac{\mathrm{d}}{\mathrm{d}t}w(t) < 0$. Taking derivative w.r.t. t we obtain:

$$\frac{\mathrm{d}w}{\mathrm{d}t} = \int_0^\ell 2uu_t \, dx = 2\int_0^\ell uu_{xx} \, dx$$

where the last equality is obatined by performing substitution by the given diffusion equation $u_t = u_x x$. Integration by parts give us:

$$\frac{\mathrm{d}w}{\mathrm{d}t} = 2[uu_x]_0^\ell - 2\int_0^\ell u_x^2\ dx = -2\int_0^\ell u_x^2\ dx$$

where the last equality is due to the given boundary conditions $u(0,t) = u(\ell,t) = 0$. Clealry $-2\int_0^\ell u_x^2\ dx$ is non-positive. We claim that it is non-zero. If not, then $u_x = 0$ implies that u is a constant. Since $u(0,t) = 0$, so u is a zero function. But we have shown that $u(x,t) > 0$, thus contradiction, implying that the integral is non-zero. Therefore

$$\frac{\mathrm{d}w}{\mathrm{d}t} = -2\int_0^\ell u_x^2\ dx < 0$$

We now turn to study stability. The general idea of stability is that, if in a system, "close" initial data implies "close" solutions, where the closeness can be measured by the L^2-distance. We first introduce the notion of distance between functions:

Definition 2.3.5 (L^2-distance)**.** Suppose given two functions f, g. The L^2-distance between f and g is defined to be

$$\mathrm{dist}(f,g) = \left(\int_0^\ell [f(x) - g(x)]^2\ dx\right)^{\frac{1}{2}}$$

Again the setup is similar: suppose given diffusion equation $u_t - ku_{xx} = f(x,t)$ where $0 < x < \ell$ and $t > 0$, accompanied with initial boundary conditions:

$$u(x,0) = \phi(x),\ 0 < x < \ell \quad \text{and} \quad u(0,t) = g(t),\ u(\ell,t) = h(t)$$

Let $u_1(x,t)$ and $u_2(x,t)$ be two solutions generated by the equation with initial values $u_1(x,0) = \phi_1(x)$ and $u_2(x,0) = \phi_2(x)$ respectively.

Notice that $w := u_1 - u_2$ satisfies the diffusion equation $w_t - kw_{xx} = 0$ with initial boundary conditions:

$$w(x,0) = \phi_1(x) - \phi_2(x),\ 0 < x < \ell \quad \text{and} \quad w(0,t) = 0,\ w(\ell,t) = 0$$

Back in Example 2.3.4, we have shown that $\int_0^\ell w^2\ dx$ is a strictly decreasing function of t, thus

$$\int_0^\ell (u_1(x,t) - u_2(x,t))^2\ dx = \int_0^\ell w^2(x,t)\ dx \le \int_0^\ell w^2(x,0)\ dx = \int_0^\ell (\phi_1(x) - \phi_2(x))^2\ dx$$

By the definition of L^2-distance, we see that the above inequality implies

$$\mathrm{dist}(u_1,u_2) \le \mathrm{dist}(\phi_1,\phi_2)$$

Thus the system is stable: if ϕ_1 and ϕ_2 are very close, then their L^2-distance will be small, and thus the L^2-distance of u_1 and u_2 is small, suggesting that the two solutions are close.

Remark 2.3.6 (Alternative formulation of Stability)**.** We can also apply Maximum Principle and Minimum Principle to show stability. Recall that the boundary conditions for $w = u_1 - u_2$ are

$$w(x,0) = \phi_1(x) - \phi_2(x),\ 0 < x < \ell \quad \text{and} \quad w(0,t) = w(\ell,t) = 0$$

So, by Maximum Principle we see

$$w(x,t) \le \max(\phi_1 - \phi_2, 0)$$

Similarly Minimum Principle implies

$$w(x,t) \ge \min(\phi_1 - \phi_2, 0)$$

Together we have $\min(\phi_1 - \phi_2, 0) \le w(x,t) \le \max(\phi_1 - \phi_2, 0)$. Again, we see stability here: as ϕ_1 approaches ϕ_2, the difference $w = u_1 - u_2$ approaches to 0.

Recall that our goal is to solve the Cauchy problem on $\mathbb{R}$:

$$\begin{cases} u_t & = ku_x x, \ x \in \mathbb{R}, \ t > 0 \\ u(x,0) & = \phi(x) \end{cases} \tag{2.3}$$

We sketch the strategy to obtain the solution: we first solve the equation $u(x,0) = \phi(x)$ for a specific $\phi(x)$ of the form

$$\phi(x) = \begin{cases} 1, \ x > 0 \\ 0, \ x < 0 \end{cases}$$

Then we build the solution for general $\phi(x)$ from this particular one. Before starting, we list some properties that will be helpful later:

Proposition 2.3.7.

1. *[Spatial translations] Given a fixed y, the translate $u(x-y,t)$ of any solution $u(x,t)$ is another solution.*

2. *[Dilation (scaling)] Given constant $a > 0$, the dilation $u(\sqrt{a}x, at)$ of any solution $u(x,t)$ is another solution.*

3. *[Differentiation] Any partial derivative of a solution is again a solution.*

4. *[Linear combinations] If $u_1, \ldots, u_n$ are solutions of (2.3), then so is $u = c_1u_1 + \cdots + c_nu_n$ for any constants $c_1, \ldots, c_n$.*

5. *[Convolution invariance] If $S(x,t)$ is a solution of (2.3), then so is*

$$v(x,t) = \int_{-\infty}^{\infty} S(x-y,t)g(y)dy =: S(\cdot,t) * g$$

for any function g.

Theorem 2.3.8 (Solution formula for the diffusion equation)**.** *The problem*

$$\begin{cases} u_t & = ku_{xx}, \ x \in \mathbb{R}, \ t > 0 \\ u(x,0) & = \phi(x) \end{cases}$$

has solution

$$u(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} \exp\left(-\frac{(x-y)^2}{4kt}\right) \phi(y) \ dy$$

Proof. Take the Heaviside step function

$$H(x) = \begin{cases} 1, \ x > 0 \\ 0, \ x < 0 \end{cases}$$

and consider the IVP with Heaviside step function as initial value

$$\begin{cases} Q_t & = kQ_{xx}, \ x \in \mathbb{R}, \ t > 0 \\ Q(x,0) & = H(x) \end{cases}$$

If $Q(x,t)$ is a solution, then by Dilation property $Q(\sqrt{a}x, at)$ also solves the diffusion equation $u_t - ku_{xx} = 0$, but we do not know if it solves the specific IVP with Heaviside step function, thus checking is required. Note by assumption $Q(x,0) = H(x)$, and thus $Q(\sqrt{a}x, 0) = H(\sqrt{a}x)$, and it is easy to show that $H(\sqrt{a}x) = H(x)$. Thus we can conclude that $Q(\sqrt{a}x, at)$ also solves the IVP with Heaviside step function. We have showned previously that the solution of diffusion equation is unique, thus $Q(\sqrt{a}x, at) = Q(x,t)$ for all $x \in \mathbb{R}$ and $t > 0$. This says that Q is invariant under dilation.

Fixed (x,t) and let $a = 1/t$, then

$$Q(x,t) = Q(\sqrt{a}x, at) = Q\left(\sqrt{\frac{1}{t}}x, \frac{1}{t}t\right) = Q\left(\sqrt{\frac{1}{t}}x, 1\right)$$

This shows that the value of Q only depends on $\frac{x}{\sqrt{t}}$. So, we shall look for $Q(x,t)$ of the special form

$$Q(x,t) = g(p) \quad \text{where} \quad p = \frac{x}{\sqrt{4kt}}$$

and g is some function of one variable. The $\sqrt{4k}$ is included for later convenience. With this, we can rewrite the IVP into an ODE of g using chain rule:

$$\begin{aligned}
Q_t &= \frac{\mathrm{d}g}{\mathrm{d}p}\frac{\partial p}{\partial t} = -\frac{1}{2t}\frac{x}{\sqrt{4kt}}g'(p) = -\frac{1}{2t}pg'(p) \\
Q_x &= \frac{\mathrm{d}g}{\mathrm{d}p}\frac{\partial p}{\partial x} = \frac{1}{\sqrt{4kt}}g'(p) \\
Q_{xx} &= \frac{\mathrm{d}Q_x}{\mathrm{d}p}\frac{\partial p}{\partial x} = \frac{1}{4kt}g''(p)
\end{aligned}$$

Since Q is a solution, we must have

$$0 = Q_t - kQ_{xx} = \frac{1}{t}\left(-\frac{1}{2}pg'(p) - \frac{1}{4}g''(p)\right)$$

Since $t > 0$ so we get $g''(p) + 2pg'(p) = 0$. This ODE can be solved as follows:

$$\begin{aligned}
\frac{g''}{g'} = -2p &\implies \ln|g'| = -p^2 + c_1 \\
&\implies g' = \pm e^{c_1}e^{-p^2} = C_1 e^{-p^2} \\
&\implies g(p) = C_1 \int e^{-p^2}\ dp + C_2
\end{aligned}$$

Thus we have the solution for $Q_t - kQ_{xx} = 0$ with $p = \frac{x}{\sqrt{4kt}}$

$$Q(x,t) = g(p) = C_1 \int e^{-p^2}\ dp + C_2 = C_1 \int_0^p e^{-s^2}\ dx + C_2$$

We now determine C_1 and C_2. By initial condition $Q(x,0) = H(x)$ of the IVP:

$$\text{if } x > 0, \quad 1 = \lim_{t\to 0^+} Q = C_1 \int_0^\infty e^{-s^2}\ dx + C_2 = C_1\frac{\sqrt{\pi}}{2} + C_2$$

$$\text{if } x < 0, \quad 0 = \lim_{t\to 0^+} Q = C_1 \int_0^\infty e^{-s^2}\ dx + C_2 = -C_1\frac{\sqrt{\pi}}{2} + C_2$$

where we have applied the well-known integral formula $\int_0^\infty e^{-s^2}\ ds = \frac{\sqrt{\pi}}{2}$. Solving out we get

$$C_1 = \frac{1}{\sqrt{\pi}} \quad \text{and} \quad C_2 = \frac{1}{2}$$

Together we get

$$Q(x,t) = \frac{1}{\sqrt{\pi}}\int_0^{x/\sqrt{4kt}} e^{-s^2}\ dx + \frac{1}{2}$$

We can finally solve the general IVP. Define

$$S(x,t) = \frac{\partial Q}{\partial x} = \frac{1}{\sqrt{4k\pi t}}\exp\left(-\frac{x^2}{4kt}\right)$$

By the Differentiation property mentioned above $S(x,t)$ is a solution to $u_t - ku_{xx} = 0$. By the Convolution Invariance property, the following is also a solution:

$$u(x,t) = \int_{-\infty}^{\infty} S(x-y,t)\phi(y)\ dy,\ t > 0$$

It remains to check that $u(x,t)$ satisfies the initial condition $u(x,0) = \phi(x)$. We first rewrite u as follows:

$$\begin{aligned} u(x,t) &= \int_{-\infty}^{\infty} \frac{\partial Q}{\partial x}(x-y,t)\phi(y)\ dy \\ &= -\int_{-\infty}^{\infty} \frac{\partial}{\partial y}[Q(x-y,t)]\phi(y)\ dy \\ &= \int_{-\infty}^{\infty} Q(x-y,t)\phi'(y)\ dy - [Q(x-y,t)\phi(y)]_{y=-\infty}^{y=\infty} \end{aligned}$$

where the second last equality is due to $\dfrac{\partial Q}{\partial x} + \dfrac{\partial Q}{\partial y} = 0$, and the last equality is due to integration by parts. We assume that the limit $[Q(x-y,t)\phi(y)]_{y=-\infty}^{y=\infty}$ vanishes by assuming that $\phi(y) = 0$ for large $|y|$. Recall we want to show $u(x,0) = \phi(x)$. By using the fact that $Q(x,0) = H(x)$, we see that

$$\begin{aligned} u(x,0) &= \int_{-\infty}^{\infty} Q(x-y,0)\phi'(y)\ dy \\ &= \int_{-\infty}^{\infty} H(x-y)\phi'(y)\ dy \\ &= \int_{-\infty}^{x} \phi'(y)\ dy \\ &= \phi(x) - \phi(-\infty) \\ &= \phi(x) \end{aligned}$$

where in the last line we use our assumption that $\phi(-\infty) = 0$. This shows that the solution $u(x,t)$ satisfies the initial condition, thus solves the IVP. □

Due to the wide usage of the function $S(x,t)$, we shall formally define it:

Definition 2.3.9 (Gaussian kernel)**.** The function

$$S(x,t) = \frac{1}{\sqrt{4k\pi t}} \exp\left(-\frac{x^2}{4kt}\right)$$

is known as *Gaussian kernel, fundamental solution, source function, Green's function,* or *propagator* of the heat equation.

It is direct to spot the following statement:

Proposition 2.3.10 (Evenness and Positivity of Gaussian kernel)**.** *Gaussian kernel $S(x,t)$ is an even function of x, and it is always positive.*

Essentially, the solution $u(x,t)$ is essentially a convolution of $S(x,t)$ with the initial value $\phi(x)$, which can be seen as a *weighting function* that 'distribute' $\phi(y)$:

$$u(x,t) = \int_{-\infty}^{\infty} S(x-y,t)\phi(y)\ dy \simeq \sum_j S(x-y_j,t)\phi(y_j)\triangle y_j$$

where $\{y_j\}$ are some sampling points. This can be interpretated as follow:

> The source function $S(x-y,t)$ weights the contribution of $\phi(y)$ according to the distance of y from x and the elapsed time t. In other words, if point y is closer to x, then it has a bigger weight $S(x-y,t)$, and vice versa.

We now show that the parity of the solution $u(x,t)$ is the same as the parity of the initial value $\phi(x)$.

Lemma 2.3.11 (Parity of solution for diffusion equation)**.** *If the initial value $\phi(x)$ is even (odd), then the solution $u(x,t)$ is also even (odd) in x.*

Proof. First suppose that $\phi(x)$ is even, i.e. $\phi(-x) = \phi(x)$ for all x. By previous theorem we have

$$\begin{aligned} u(-x,t) &= \int_{-\infty}^{\infty} S(-x-y,t)\phi(y)\ dy \\ &= -\int_{-\infty}^{\infty} S(-x+z,t)\phi(z)\ dz \quad \text{where } z := -y \\ &= \int_{-\infty}^{\infty} S(-x+z,t)\phi(z)\ dz \quad \text{since } \phi \text{ is even} \\ &= \int_{-\infty}^{\infty} S(x-z,t)\phi(z)\ dz \quad \text{since } S \text{ is even by Proposition 2.3.10} \end{aligned}$$

We see that z is a dummy variable, so we can replace it by y to get that $u(-x,t) = u(x,t)$, which shows that u is even.

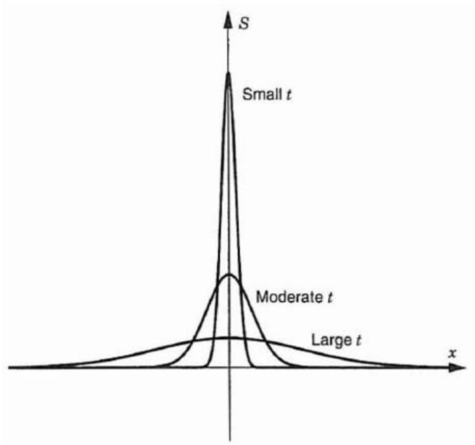
Next, suppose that $\phi(x)$ is odd, i.e. $\phi(-x) = -\phi(x)$ for all x. Similarly, we start from

$$\begin{aligned} u(-x,t) &= \int_{-\infty}^{\infty} S(-x-y,t)\phi(y)\ dy \\ &= -\int_{-\infty}^{\infty} S(-x+z,t)\phi(z)\ dz \quad \text{where } z := -y \\ &= -\int_{-\infty}^{\infty} S(x-z,t)\phi(z)\ dz \quad \text{since } S \text{ is even by Proposition 2.3.10} \end{aligned}$$

Again, we can replace z by y to get that $u(-x,t) = -u(x,t)$, which shows that u is odd. □

Remark 2.3.12.

- The name 'propagator' can be understood as propagating the initial data to the later time, giving the solution at any time $t > 0$.
- When t is large, $S(x,t)$ is very spread out; it is a thin tall spike of height $\frac{1}{\sqrt{4k\pi t}}$ when t is small.



- The area under the curve of $S(x,t)$ is 1 for any $t \geq 0$:

$$\begin{aligned} \int_{-\infty}^{\infty} S(x,t)\ dx &= \frac{1}{\sqrt{4k\pi t}} \int_{-\infty}^{\infty} \exp\left(-\frac{x^2}{4kt}\right)\ dx \\ &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-q^2}\ dq \quad \text{where we substitute } q = \frac{x}{\sqrt{4kt}} \\ &= \frac{1}{\sqrt{\pi}}\sqrt{\pi} = 1 \end{aligned}$$

- The name 'fundamental solution' is given since $S(x,t)$ is the solution of the IVP:

$$\begin{cases} u_t = ku_{xx}, \ x \in \mathbb{R}, \ t > 0 \\ u(x,0) = \delta(x) \end{cases}$$

  where δ is the Dirac delta function. This follows from the fact that Q solves the IVP where $u(x,0) = H(x)$, thus $S(x,0) = Q_x(x,0) = H'(x) = \delta(x)$.

- Regardless of the smoothness of the initial condition $\phi(x)$, the solution $u(x,t)$ is sufficiently smooth for $t > 0$ and any x, since all partial derivatives of u exists.

Together, we know when t is small, the source function $S(x,t)$ is a spike that the formula exaggerates the value of ϕ near x. Thus as t increases, the function $S(x,t)$ will spread out the initial value $\phi(x)$. This makes sense in the setting of trasfer of heat: as time increases, the heat is transfered and slowly became homogenous throughout the whole body.

It is usually impossible to evaluate the solution $u(x,t)$ completely in terms of elementary functions. However, for some particular initial data $\phi(x)$, the solutions are expressible in terms of the *error function.*

Definition 2.3.13 (Error function)**.** The error function is defined to be

$$\mathrm{Erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-p^2} \, dp$$

A direct computataion tells that $\mathrm{Erf}(0) = 0$, and

$$\lim_{x\to\infty} \mathrm{Erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-p^2} \, dp = \frac{2}{\sqrt{\pi}} \frac{\sqrt{\pi}}{2} = 1$$

We have in fact seen the error function previously. Recall that in the proof of Theorem 2.3.8, we defined

$$Q(x,t) = \frac{1}{2} + \frac{1}{\sqrt{\pi}} \int_0^{x/\sqrt{4kt}} e^{-s^2} \, dx$$

In fact we can write it as

$$Q(x,t) = \frac{1}{2} + \frac{1}{2} \mathrm{Erf}\left(\frac{x}{\sqrt{4kt}}\right)$$

Example 2.3.14. Consider the IVP

$$\begin{cases} u_t = ku_{xx}, \ x \in \mathbb{R} \\ u(x,0) = \phi(x) = \begin{cases} 100 & , \ x > 0 \\ 0 & , \ x < 0 \end{cases} \end{cases}$$

We will evaluate its solution using the error function. By the solution formula we have

$$\begin{aligned} u(x,t) &= \frac{1}{\sqrt{4k\pi t}} \int_0^\infty \exp\left(-\frac{(x-y)^2}{4kt}\right) \, dy \\ &= \frac{100}{\sqrt{\pi}} \int_{-x/\sqrt{4kt}}^\infty e^{-q^2} \, dq \quad \text{where we substitue } q := \frac{y-x}{\sqrt{4kt}} \\ &= \frac{100}{\sqrt{\pi}} \int_0^\infty e^{-q^2} \, dq + \frac{100}{\sqrt{\pi}} \int_{-x/\sqrt{4kt}}^0 e^{-q^2} \, dq \\ &= \frac{100}{\sqrt{\pi}} \frac{\sqrt{\pi}}{2} + \frac{100}{\sqrt{\pi}} \int_0^{x/\sqrt{4kt}} e^{-q^2} \, dq \\ &= 50 + 50 \, \mathrm{Erf}\left(\frac{x}{\sqrt{4kt}}\right) \end{aligned}$$

where the second last equality is utilizing the fact that e^{-q^2} is an even function.

2.4 Comparison of Waves and Diffusions

We have learned that waves are essentially transporting the initial data in both directions at a finite speed. On the other hand, diffusions are spreading the initial datas in a smooth fashion and gradually disapper. Also, for wave equation, the energy is preserved as a constant, while in diffusion equation the energy decays to zero. They are also different from the perspective of well-posedness: wave equation is well-posed for all t, but diffusion equation is not for $t < 0$.

We end this chapter with an example demonstrating that Maximum Principle does not hold for the wave equation:

Example 2.4.1. Consider the wave equation $u_{tt} = u_{xx}$ with initial condition

$$u(x,0) = 0 \quad \text{and} \quad u_t(x,0) = \psi(x) = \begin{cases} 1, |x| < 1 \\ 0, |x| > 1 \end{cases}$$

Suppose that the Maximum Principle holds for wave equation. Since in this case there is no lateral boundary, by the principle the maximum must be attained at $t = 0$. However, when $t = 0$, the initial condition gives $u(x,0) = 0$, and based on d'Alembert formula the solution is positive for $t > 0$. This shows that maximum principle does not hold for wave equation.

Chapter 3

Reflection and Sources

In this chapter, we will be solving diffusion equations and wave equations on the half-line $\mathbb{R}^+$.

3.1 Diffusion on the Half-Line

We first define a specific type of diffusion problem on the half-line:

Definition 3.1.1 (IBVP with Dirichlet Boundary Condition of diffusion equation)**.** The initial/boundary value problem (IBVP) with a Dirichlet boundary condition at the endpoint $x = 0$ is given by:

$$\begin{cases} v_t = kv_{xx}, & x \in \mathbb{R}^+, t > 0 \\ v(0,t) = 0, & t > 0 \\ v(x,0) = \phi(x), & x > 0 \end{cases}$$

where we assume $\phi(0) = 0$.

Here $v(x,0) = \phi(x)$ is called the initial condition, $v(0,t) = 0$ is called the boundary condition, and $\phi(0) = 0$ is called the consistent condition. Note that if a solution exists, it is unique by the Maximum Principle.

Remark 3.1.2 (Physical interpretation of IBVP with Dirichlet Boundary Condition of diffusion equation)**.** The setup for IBVP with Dirichlet Boundary Condidtion of diffusion question can be interpreted as having an infinite rod with the initial temperature given by $\phi(x)$, and the endpoint $x = 0$ is kept at zero temperature for all time. The question is to find the temperature distribution $v(x,t)$ for $x > 0$ and $t > 0$.

A strategy to solve the above question is to extend the initial condition ϕ to the whole line, so that we can use the solution formula for the diffusion equation on the whole line given in Theorem 2.3.8.

Definition 3.1.3 (Odd extension of a function)**.** The odd extension of a function ϕ defined on $\mathbb{R}^+$ is given by:

$$\psi(x) = \begin{cases} \phi(x) & , x > 0 \\ 0 & , x = 0 \\ -\phi(-x) & , x < 0 \end{cases}$$

It is immediate that $\psi(x)$ is an odd function.

We can now solve the IBVP with Dirichlet Boundary Condition of diffusion equation by using the odd extension of the initial condition:

Lemma 3.1.4. *The solution to the IBVP with Dirichlet Boundary Condition of diffusion equation is given by:*

$$v(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_0^\infty \left(\exp\left(-\frac{(x-y)^2}{4kt} \right) - \exp\left(-\frac{(x+y)^2}{4kt} \right) \right) \phi(y) dy$$

Proof. Let ψ be the odd extension of ϕ. Consider the Cauchy problem:

$$\begin{cases} u_t - ku_{xx} = 0 & , \ x \in \mathbb{R}, \ t > 0 \\ u(x,0) = \psi(x) & , \ x \in \mathbb{R} \end{cases}$$

By Theorem 2.3.8, the solution to the above Cauchy problem is given by:

$$u(x,t) = \int_{-\infty}^{\infty} S(x-y,t)\psi(y)\ dy = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} \exp\left(-\frac{(x-y)^2}{4kt}\right)\psi(y)\ dy$$

Next, define $v(x,t) = u(x,t)|_{x\geq 0}$. We claim that v is the solution to the IBVP with Dirichlet Boundary Condition of diffusion equation. It is clear that v satisfies the diffusion equation on the positive half-line. Moreover, by definition chasing, we see

$$v(x,0) = u(x,0)|_{x>0} = \psi(x)|_{x>0} = \phi(x)$$

This shows that v satisfies the initial condition. Finally, we check the boundary condition. We first check that for any odd function f it must satisfies $f(0) = 0$:

if f is odd, then $f(0) = f(-0) = -f(0)$, which implies $f(0) = 0$.

Recall in Lemma 2.3.11 we have shown that if the initial condition is odd, then the solution to the diffusion equation is also odd. Since ψ is odd, u is also odd by Lemma 2.3.11. Therefore, we have

$$v(0,t) = u(0,t) = 0$$

where the last equality follows from the fact that u is an odd function. Lastly, the formula for v can be obtained as follows: first we rewrite u as

$$\begin{aligned} u(x,t) &= \int_{-\infty}^{\infty} S(x-y,t)\psi(y)\ dy \\ &= \int_{-\infty}^{0} S(x-y,t)\psi(y)\ dy + \int_{0}^{\infty} S(x-y,t)\psi(y)\ dy \\ &= -\int_{-\infty}^{0} S(x-y,t)\phi(-y)\ dy + \int_{0}^{\infty} S(x-y,t)\phi(y)\ dy \quad \text{since } \psi \text{ is the odd extension of } \phi \\ &= -\int_{-\infty}^{0} S(x+z,t)\phi(z)\ dz + \int_{0}^{\infty} S(x-y,t)\phi(y)\ dy \quad \text{where } z := -y \\ &= \int_{0}^{\infty} S(x-y,t)\phi(y)\ dy - \int_{0}^{\infty} S(x+y,t)\phi(y)\ dy \\ &= \int_{0}^{\infty} \left(S(x-y,t) - S(x+y,t)\right)\phi(y)\ dy \end{aligned}$$

Restricting to $x > 0$ gives the desired formula for v. □

Here is a different type of diffusion problem on the half-line:

Definition 3.1.5 (IBVP with Neumann Boundary Condition of diffusion equation)**.** The initial/boundary value problem (IBVP) with a Neumann boundary condition at the endpoint $x = 0$ is given by:

$$\begin{cases} w_t = kw_{xx} & , \ x \in \mathbb{R}^+, \ t > 0 \\ w_x(0,t) = 0 & , \ t > 0 \\ w(x,0) = \phi(x) & , \ x > 0 \end{cases}$$

Again, similarly to the previous case, the above problem must have a unique solution if it exists, by the Maximum Principle. The strategy to solve the above question is also similar: extending the initial data $\phi(x)$ to the entire line, and solve the diffusion equation on the whole line. However, the extension of ϕ is different from the previous case:

Definition 3.1.6 (Even extension of a function)**.** The even extension of a function ϕ defined on $\mathbb{R}^+$ is given by:

$$\varphi(x) = \begin{cases} \phi(x) & , \ x \geq 0 \\ \phi(-x) & , \ x \leq 0 \end{cases}$$

Note that if $\phi(x)$ is an even function, then taking differentiation from the identity $\phi(-x) = \phi(x)$ gives

$$-\phi'(-x) = \phi'(x)$$

This implies that ϕ' is an odd function, and thus $\phi'(0) = 0$. It is now clear that why an even extension of the initial data ϕ is needed instead of an odd extension: we want the solution to satisfy the Neumann boundary condition, which requires that $\phi'(0) = 0$.

Lemma 3.1.7. *The solution to the IBVP with Neumann Boundary Condition of diffusion equation is given by:*

$$w(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_0^\infty \left(\exp\left(-\frac{(x-y)^2}{4kt} \right) + \exp\left(-\frac{(x+y)^2}{4kt} \right) \right) \phi(y) dy$$

Proof. Let φ be the even extension of ϕ. Consider the Cauchy problem:

$$\begin{cases} u_t - ku_{xx} = 0 & , \ x \in \mathbb{R}, \ t > 0 \\ u(x,0) = \varphi(x) & , \ x \in \mathbb{R} \end{cases}$$

By Theorem 2.3.8, the solution to the above Cauchy problem is given by:

$$u(x,t) = \int_{-\infty}^{\infty} S(x-y,t)\varphi(y) \ dy = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} \exp\left(-\frac{(x-y)^2}{4kt} \right) \varphi(y) \ dy$$

Next, define $v(x,t) = u(x,t)|_{x \geq 0}$. We claim that v is the solution to the IBVP with Neumann Boundary Condition of diffusion equation. It is clear that v satisfies the diffusion equation on the positive half-line. Moreover, by definition chasing, we see

$$v(x,0) = u(x,0)|_{x>0} = \varphi(x)|_{x>0} = \phi(x)$$

This shows that v satisfies the initial condition. Finally, we check the boundary condition. Recall in Lemma 2.3.11 we have shown that if the initial condition is even, then the solution to the diffusion equation is also even. Since φ is even, u is also even by Lemma 2.3.11. We have shown previously that if f is an even function, then $f'(0) = 0$, thus

$$v_x(0,t) = u_x(0,t) = 0$$

Lastly, the formula for v can be obtained as follows: first we rewrite u as

$$\begin{aligned}
u(x,t) &= \int_{-\infty}^{\infty} S(x-y,t)\varphi(y) \ dy \\
&= \int_{-\infty}^{0} S(x-y,t)\varphi(y) \ dy + \int_0^\infty S(x-y,t)\varphi(y) \ dy \\
&= \int_{-\infty}^{0} S(x-y,t)\phi(-y) \ dy + \int_0^\infty S(x-y,t)\phi(y) \ dy \quad \text{since } \varphi \text{ is the even extension of } \phi \\
&= \int_0^\infty S(x+z,t)\phi(z) \ dz + \int_0^\infty S(x-y,t)\phi(y) \ dy \quad \text{where } z := -y \\
&= \int_0^\infty S(x+y,t)\phi(y) \ dy + \int_0^\infty S(x-y,t)\phi(y) \ dy \\
&= \int_0^\infty \left(S(x+y,t) + S(x-y,t) \right) \phi(y) \ dy
\end{aligned}$$

Restricting to $x > 0$ gives the desired formula for v. $\square$

3.2 Reflection of Waves

In this section, we perform a similar analysis for the wave equation on the half-line. We first define the following IBVP for the wave equation:

Definition 3.2.1 (IBVP with Dirichlet Boundary Condition of wave equation)**.** The initial/boundary value problem (IBVP) with a Dirichlet boundary condition at the endpoint $x = 0$ is given by:

$$\begin{cases} v_{tt} = c^2 v_{xx} & , \ x \in \mathbb{R}^+, t > 0 \\ v(0,t) = 0 & , \ t > 0 \\ v(x,0) = \phi(x) & , \ x > 0 \\ v_t(x,0) = \psi(x) & , \ x > 0 \end{cases}$$

where we assume $\phi(0) = 0$.

If the solution exists, then it must be unique by the energy method. Similarly, the strategy is to entend the nitial data ϕ and ψ to the whole line, and solve the wave equation on the whole line. We shall consider its odd extension, as we are dealing with the Dirichlet form of the IBVP.

Lemma 3.2.2. *The solution to the IBVP with Dirichlet Boundary Condition of wave equation is given by:*

$$v(x,t) = \begin{cases} \frac{1}{2}\left(\phi(x+ct) + \phi(x-ct)\right) + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)\ ds & , x > ct \\ \frac{1}{2}\left(\phi(x+ct) - \phi(ct-x)\right) + \frac{1}{2c}\int_{ct-x}^{ct+x} \psi(s)\ ds & , 0 < x < ct \end{cases}$$

Proof. Let $\tilde{\phi}$ and $\tilde{\psi}$ be the odd extensions of ϕ and ψ respectively. Consider the Cauchy problem:

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 & , \ x \in \mathbb{R}, t > 0 \\ u(x,0) = \tilde{\phi}(x) & , \ x \in \mathbb{R} \\ u_t(x,0) = \tilde{\psi}(x) & , \ x \in \mathbb{R} \end{cases}$$

We define the restriction of the solution $u(x,t)$ to the positive half-line as

$$v(x,t) := u(x,t)|_{x \geq 0}$$

and we claim that v is the solution to the IBVP with Dirichlet Boundary Condition of wave equation. It is clear that v satisfies the wave equation on the positive half-line.

We verify that v satisfies the initial/boundary conditions. First, since $\tilde{\phi}$ and $\tilde{\psi}$ are odd functions, the solution u to the above Cauchy problem is also an odd function, and thus for all $t > 0$ we have

$$u(0,t) = 0 \implies v(0,t) = 0$$

Next, the initial condition of the Cauchy problem $u(x,0) = \tilde{\phi}(x)$ implies that for all $x > 0$ we have

$$v(x,0) = \phi(x)$$

Similarly the initial condition of the Cauchy problem $u_t(x,0) = \tilde{\psi}(x)$ implies that for all $x > 0$ we have

$$v_t(x,0) = \psi(x)$$

This shows that v is indeed the required solution. It remains to derive its formula. By D'Alembert's formula for the Cauchy problem of wave equation, we have that for $x \in \mathbb{R}$ that

$$u(x,t) = \frac{1}{2}\left(\tilde{\phi}\left(x+ct\right) + \tilde{\phi}\left(x-ct\right)\right) + \frac{1}{2c}\int_{x-ct}^{x+ct} \tilde{\psi}\left(s\right)\ ds$$

We have to restrict the solution to $x \geq 0$ to derive the formula for v. In particular, we have to decide what value of $\tilde{\phi}$ and $\tilde{\psi}$ to use, which depends on the sign of $x + ct$ and $x - ct$. First note that in the IBVP we always have $x \geq 0$ and $t > 0$, since $c > 0$ and thus $x + ct > 0$, thus

$$\tilde{\phi}\left(x+ct\right) = \phi\left(x+ct\right) \quad \text{and} \quad \tilde{\psi}\left(x+ct\right) = \psi\left(x+ct\right)$$

It now remains to determine the sign of $x - ct$. First, if $x - ct > 0$, then we have

$$v(x,t) = \frac{1}{2}\left(\phi\left(x+ct\right) + \phi\left(x-ct\right)\right) + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi\left(s\right)\ ds$$

Next, if $x - ct < 0$, then we have

$$\begin{aligned}
v(x,t) &= \frac{1}{2}\left(\tilde{\phi}\left(x+ct\right) + \tilde{\phi}\left(x-ct\right)\right) + \frac{1}{2c}\int_{x-ct}^{x+ct} \tilde{\psi}\left(s\right)\ ds \\
&= \frac{1}{2}\left(\phi\left(x+ct\right) - \phi\left(ct-x\right)\right) + \frac{1}{2c}\left(\int_{x-ct}^{0} \tilde{\psi}\left(s\right)\ ds + \int_{0}^{x+ct} \tilde{\psi}\left(s\right)\ ds\right) \\
&= \frac{1}{2}\left(\phi\left(x+ct\right) - \phi\left(ct-x\right)\right) + \frac{1}{2c}\left(\int_{x-ct}^{0} -\psi\left(-s\right)\ ds + \int_{0}^{x+ct} \psi\left(s\right)\ ds\right) \\
&= \frac{1}{2}\left(\phi\left(x+ct\right) - \phi\left(ct-x\right)\right) + \frac{1}{2c}\left(\int_{ct-x}^{0} \psi\left(y\right)\ dy + \int_{0}^{x+ct} \psi\left(s\right)\ ds\right) \text{ where } y := -s \\
&= \frac{1}{2}\left(\phi\left(x+ct\right) - \phi\left(ct-x\right)\right) + \frac{1}{2c}\int_{ct-x}^{ct+x} \psi\left(s\right)\ ds
\end{aligned}$$

This completes the proof. □

When a wave is reflected, it is possible that the reflection of the wave has different sign from the original one. If peaks are reflected to peaks and troughs are reflected to troughs, then both has the same size. On the other hand, if peaks are reflected to troughs and troughs are reflected to peaks, then both has the same size but different sign.

Also, recall from d'Alambert Formula that on $\mathbb{R}$, every solution is $u(x,t) = F(x-ct) + G(x+ct)$, where F and G are determined by the initial conditions. In particular, F is the right-moving wave and G is the left-moving wave.

Remark 3.2.3 (Physical Interpretation and the Reflection of waves at the boundary)**.** The setup for IBVP with Dirichlet Boundary Condidtion of wave equation can be interpreted as an infinite string with:

- initial displacement given by $\phi(x)$
- initial velocity given by $\psi(x)$
- the endpoint $x = 0$ is fixed for all time (this corresponds to $v(0,t) = 0$)

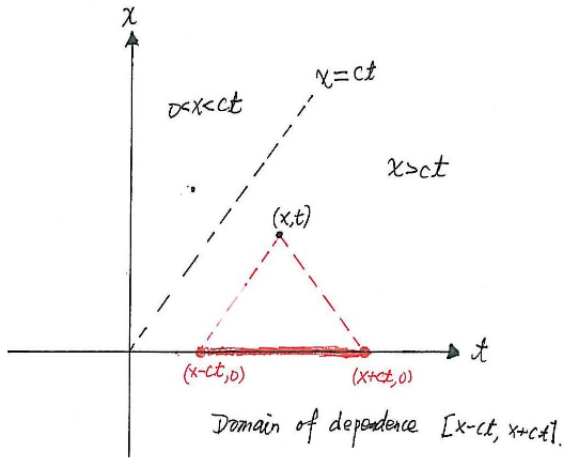
Note that the wave travelling to the left $\phi(x+ct)$ is reflected at the fixed boundary $x = 0$, which we have to decide whether the reflection changes the sign of the wave or not. At $x = 0$, by the d'Alambert formula we get

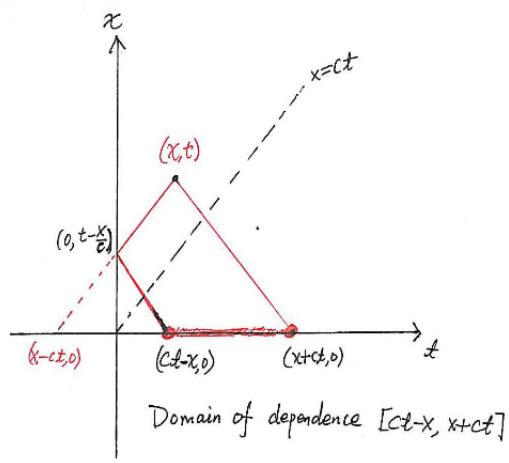
$$u(0,t) = F(-ct) + G(ct) = 0$$

This implies that $G(ct) = -F(-ct)$, which means that the reflection of the wave changes the sign of the wave.

Physically, this implies that if the starting point is close enough to the boundary, the opposite-signed reflected wave can chase up the original waves, and the cancellation occurs due to the opposite signs. Of course, this will not happen if the starting point is far enough from the boundary.

In particular, we have seen that 'close enough' means $0 < x < ct$, and 'far enough' means $x > ct$. Recall c is the speed of the wave, so the wave has travelled a distance of ct after time t.





Definition 3.2.4 (IBVP with Neumann Boundary Condition of wave equation)**.** The initial/boundary value problem (IBVP) with a Neumann boundary condition at the endpoint $x = 0$ is given by:

$$\begin{cases} w_{tt} = c^2 w_{xx} & , \ x \in \mathbb{R}^+, \ t > 0 \\ w_x(0,t) = 0 & , \ t > 0 \\ w(x,0) = \phi(x) & , \ x \in \mathbb{R}^+ \\ w_t(x,0) = \psi(x) & , \ x \in \mathbb{R}^+ \end{cases}$$

Again, we use the reflection method to solve the above question. Even extensions are used in this case, as we are dealing with the Neumann form of the IBVP.

Lemma 3.2.5. *The solution to the IBVP with Neumann Boundary Condition of wave equation is given by:*

$$w(x,t) = \begin{cases} \dfrac{\phi(x+ct) + \phi(x-ct)}{2} + \dfrac{1}{2c} \displaystyle\int_{x-ct}^{x+ct} \psi(s) \ ds & , x > ct \\ \dfrac{\phi(x+ct) + \phi(ct-x)}{2} + \dfrac{1}{2c} \left(\displaystyle\int_0^{ct-x} + \int_0^{x+ct} \right) \psi(s) \ ds & , 0 < x < ct \end{cases}$$

Proof. Define $\tilde{\phi}$ and $\tilde{\psi}$ be the even extension of ϕ and ψ respectively. Consider the Cauchy problem:

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 & , \ x \in \mathbb{R}, t > 0 \\ u(x,0) = \tilde{\phi}(x) & , \ x \in \mathbb{R} \\ u_t(x,0) = \tilde{\psi}(x) & , \ x \in \mathbb{R} \end{cases}$$

We define the restriction of the solution $u(x,t)$ to the positive half-line as

$$w(x,t) := u(x,t)|_{x \geq 0}$$

and we claim that w is the solution to the IBVP with Neumann Boundary Condition of wave equation. It is clear that w satisfies the wave equation on the positive half-line.

We verify that w satisfies the initial/boundary conditions. First, since $\tilde{\phi}$ and $\tilde{\psi}$ are even functions, the solution u to the above Cauchy problem is even in x, and thus $u_x(x,t)$ is odd in x, hencefor all $t > 0$ we have

$$w_x(0,t) = u_x(0,t) = 0$$

Next, the initial condition of the Cauchy problem $u(x,0) = \tilde{\phi}(x)$ implies that for all $x > 0$ we have

$$w(x,0) = \phi(x)$$

Similarly the initial condition of the Cauchy problem $u_t(x,0) = \tilde{\psi}(x)$ implies that for all $x > 0$ we have

$$w_t(x,0) = \psi(x)$$

This shows that w is indeed the required solution. It remains to derive its formula. By D'Alembert's formula for the Cauchy problem of wave equation, we have that for $x \in \mathbb{R}$ that

$$u(x,t) = \frac{1}{2}\left(\tilde{\phi}\left(x+ct\right) + \tilde{\phi}\left(x-ct\right)\right) + \frac{1}{2c}\int_{x-ct}^{x+ct} \tilde{\psi}\left(s\right)\ ds$$

We have to restrict the solution to $x \geq 0$ to derive the formula for w. In particular, we have to decide what value of $\tilde{\phi}$ and $\tilde{\psi}$ to use, which depends on the sign of $x+ct$ and $x-ct$. First note that in the IBVP we always have $x \geq 0$ and $t > 0$, since $c > 0$ and thus $x + ct > 0$, thus

$$\tilde{\phi}\left(x+ct\right) = \phi\left(x+ct\right) \quad \text{and} \quad \tilde{\psi}\left(x+ct\right) = \psi\left(x+ct\right)$$

It now remains to determine the sign of $x - ct$. First, if $x - ct > 0$, then we have

$$w(x,t) = \frac{\phi(x+ct)+\phi(x-ct)}{2} + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)\ ds$$

Next, if $x - ct < 0$, then we have

$$\begin{aligned}
w(x,t) &= \frac{1}{2}\left(\tilde{\phi}\left(x+ct\right) + \tilde{\phi}\left(x-ct\right)\right) + \frac{1}{2c}\int_{x-ct}^{x+ct} \tilde{\psi}\left(s\right)\ ds \\
&= \frac{1}{2}\left(\phi\left(x+ct\right) + \phi\left(ct-x\right)\right) + \frac{1}{2c}\left(\int_{x-ct}^{0} + \int_{0}^{x+ct}\right)\tilde{\psi}\left(s\right)\ ds \\
&= \frac{1}{2}\left(\phi\left(x+ct\right) + \phi\left(ct-x\right)\right) + \frac{1}{2c}\left(\int_{x-ct}^{0} \psi(-s)\ ds + \int_{0}^{x+ct} \psi(s)\ ds\right) \\
&= \frac{1}{2}\left(\phi\left(x+ct\right) + \phi\left(ct-x\right)\right) + \frac{1}{2c}\left(\int_{0}^{ct-x} \psi(y)\ dy + \int_{0}^{x+ct} \psi(s)\ ds\right) \text{ where } y := -s
\end{aligned}$$

As y is just a dummy variable, we can replace it with s in the first integral. This completes the proof. □

Remark 3.2.6 (Physical Interpretation and the Reflection of Waves at the Boundary)**.** The setup for IBVP with Neumann Boundary Condidtion of wave equation can be interpreted as an infinite string with:

- initial displacement given by $\phi(x)$
- initial velocity given by $\psi(x)$
- the endpoint $x = 0$ is free to move vertically and staying zero slope for all time (this corresponds to $w_x(0,t) = 0$)

Similar to the case of Dirichlet boundary condition, we have to decide whether the reflection of the wave changes the sign of the wave or not. Note that despite the end is free to move vertically, the horizontal position of the end is fixed, thus the wave is reflected at the boundary.

At $x = 0$, by the d'Alambert formula we get

$$w_x(0,t) = F'(-ct) + G'(+ct) = 0$$

This implies that $G'(+ct) = -F'(-ct)$. For convenience we write $G'(s) = -F'(-s)$ with $s := ct$. Integrating w.r.t. s gives $G(s) = F(-s) + C$. The constant C can must be 0, since we assumed finite energy. In short, this shows that the reflected wave has the same sign as the original wave.

The analysis is similar: if close enough, i.e. $0 < x < ct$, the reflected wave can chase up the original wave. Since the sign of both waves are the same, by Superposition Principle, the reflected wave can amplify the original wave. If far enough, i.e. $x > ct$, the reflected wave cannot catch up the original wave, thus no amplification occurs, and the solution is the same as the solution on the whole line.

3.3 Diffusion with a Source

In this section, we study diffusion with a source.

Definition 3.3.1 (Diffusion with a source on $\mathbb{R}$)**.** The PDE system that represents diffusion with a source on the whole line is defined as:

$$\begin{cases} u_t - ku_{xx} = f(x,t) & , \ x \in \mathbb{R}, \ t > 0 \\ u(x,0) = \phi(x) \end{cases}$$

where $f(x,t)$ and $\phi(x)$ are arbitrary given functions.

Here, the function $f(x,t)$ is called the source term. Physically, it measures the physical effect of an external heat source, giving instantaneous temperature change.

Theorem 3.3.2. *The solution to the PDE system above is*

$$u(x,t) = \int_{-\infty}^{\infty} S(x-y,t)\phi(y)\ dy + \int_0^t \int_{-\infty}^{\infty} S(x-y,t-s)f(y,s)\ dyds$$

where $S(x,t)$ is the Gaussian kernel.

Proof. We show that it is indeed an solution by direct substituition. Taking differentiation w.r.t. t, we see

$$\begin{aligned} &\frac{\partial}{\partial t}u(x,t) \\ &= \frac{\partial}{\partial t}\int_{-\infty}^{\infty} S(x-y,t)\phi(y)\ dy + \frac{\partial}{\partial t}\int_0^t \int_{-\infty}^{\infty} S(x-y,t-s)f(y,s)\ dyds \\ &= \int_{-\infty}^{\infty} \frac{\partial}{\partial t}S(x-y,t)\phi(y)\ dy + \int_{-\infty}^{\infty} S(x-y,0)f(y,t)\ dy + \int_0^t \int_{-\infty}^{\infty} \frac{\partial}{\partial t}S(x-y,t-s)f(y,s)\ dyds \end{aligned}$$

Recall that Gaussian kernel is the solution for the diffusion equation with Dirac delta function as the initial delta, i.e.

$$\begin{cases} S_t = kS_{xx} \\ S(x-y,0) = \delta(x-y) \end{cases}$$

We substitute $S_t = kS_{xx}$ into the above equation, giving us:

$$\begin{aligned} &\frac{\partial}{\partial t}u(x,t) \\ &= \int_{-\infty}^{\infty} k\frac{\partial^2}{\partial x^2}S(x-y,t)\phi(y)\ dy + \int_{-\infty}^{\infty} \delta(x-y)f(y,t)\ dy + \int_0^t \int_{-\infty}^{\infty} k\frac{\partial^2}{\partial x^2}S(x-y,t-s)f(y,s)\ dyds \\ &= k\frac{\partial^2}{\partial x^2}\int_{-\infty}^{\infty} S(x-y,t)\phi(y)\ dy + f(y,t) + k\frac{\partial^2}{\partial x^2}\int_0^t \int_{-\infty}^{\infty} S(x-y,t-s)f(y,s)\ dyds \\ &= k\frac{\partial^2}{\partial x^2}\left(\int_{-\infty}^{\infty} S(x-y,t)\phi(y)\ dy + \int_0^t \int_{-\infty}^{\infty} S(x-y,t-s)f(y,s)\ dyds\right) + f(y,t) \\ &= k\frac{\partial^2}{\partial x^2}u(x,t) + f(x,t) \end{aligned}$$

This shows that $u(x,t)$ solves the inhomogenous diffusion equation. Next, verify the initial condition: Let $t \to 0$, the first term of the solution u becomes

$$\int_{-\infty}^{\infty} S(x-y,0)\phi(y)\ dy = \int_{-\infty}^{\infty} \delta(x-y)\phi(y)\ dy = \phi(x)$$

The second term of the solution u becomes an integral where s is evaluated from 0 to 0, thus vanishes. This proves that

$$u(x,0) = \phi(x)$$

This completes the proof. □

Remark 3.3.3. The interpretation of the first term of the solution is that the initial temperature distribution will be averaged out along the entire rod. But note that there is also an external heat source, thus it must be averaged out, not only over the entire rod, but over time as well.

We can also study the diffusion with a source on a half-line:

Definition 3.3.4 (Dirichlet diffusion with source on half-line)**.** The diffusion with a source on a half-line is modelled by the following system:

$$\begin{cases} v_t - kv_{xx} = f(x,t), \ x \in \mathbb{R}^+, \ t > 0 \\ v(0,t) = h(t) \\ v(x,0) = \phi(x) \end{cases}$$

This is a Dirichlet problem, and $h(t)$ is the boundary source.

The strategy to solve the Dirichlet problem is also the method of reflection and odd extension, but it requies homogenous boundary condition (i.e. the condition when $x = 0$ is 0). To do this, consider the substitution

$$V(x,t) := v(x,t) - h(t)$$

One can easily verify that $V(x,t)$ solves

$$\begin{cases} V_t - kV_{xx} = f(x,t) - h'(t), \ x \in \mathbb{R}^+, \ t > 0 \\ V(0,t) = \\ V(x,0) = \phi(x) - h(0) \end{cases}$$

This problem then has a homogenous boundary condition, and so the method of reflection can now be applied. Once $V(x,t)$ is solved, the solution $v(x,t)$ can be recovered by

$$v(x,t) = V(x,t) + h(t)$$

Of course, Neumann problem can also be solved:

Definition 3.3.5 (Neumann diffusion with source on half-line)**.** The diffusion with a source on a half-line is modelled by the following system:

$$\begin{cases} w_t - kw_{xx} = f(x,t), \ x \in \mathbb{R}^+, \ t > 0 \\ w_x(0,t) = h(t) \\ w(x,0) = \phi(x) \end{cases}$$

This is a Neumann problem, and $h(t)$ is the boundary source.

Similarly, define

$$W(x,t) = w(x,t) - xh(t)$$

and check that it satisfies

$$\begin{cases} W_t - kW_{xx} = f(x,t) - xh'(t), \ x \in \mathbb{R}^+, \ t > 0 \\ W_x(0,t) = 0 \\ W(x,0) = \phi(x) - xh(0) \end{cases}$$

This problem has a homogenous boundary condition, and thus the method of reflection and even extension can be used. The solution $w(x,t)$ can be recovered using

$$w(x,t) = W(x,t) + xh(t)$$

once $W(x,t)$ is solved.

3.4 Waves with a Source

We now study the same problem, except that the setting is now on wave equations:

Definition 3.4.1 (Waves with a source on $\mathbb{R}$)**.** The PDE system that represents waves with a source on the whole line is defined as:

$$\begin{cases} u_{tt} - c^2 u_{xx} = f(x,t) \quad , \ x \in \mathbb{R}, \ t > 0 \\ u(x,0) = \phi(x) \\ u_t(x,0) = \psi(x) \end{cases}$$

where $f(x), \phi(x)$ and $\psi(x)$ are arbitrary function.

Here, the function $f(x,t)$ is the source term. Physically, it measures the external force applied on the string when the string is vibrating.

Theorem 3.4.2. *The solution to the PDE system above is*

$$u(x,t) = \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(y)\ dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(y,s)\ dyds$$

Proof. Recall that the characteristic triange of the point (x_0,t_0) is the triangular region with three vertices $(x_0,t_0),(x_0-ct_0,0)$, and $(x_0+ct_0,0)$, which we denote as $\triangle$. Taking integration on f over $\triangle$ gives

$$\iint_\triangle f\ dxdt = \iint_\triangle (u_{tt} - c^2 u_{xx})\ dxdt$$

We will be applying Green's Theorem. Let $P = -c^2 u_x$ and $Q = -u_t$, then

$$P_x - Q_t = -c^2 u_{xx} + u_{tt}$$

Thus the equation above can be rewritten using Green's Theorem:

$$\iint_\triangle f\ dxdt = \int_{L_0+L_1+L_2} (-c^2 u_x\ dt - u_t\ dx)$$

where L_0, L_1, L_2 are the lines from $(x_0-ct_0,0)$ to $(x_0+ct_0,0)$,$(x_0+ct_0,0)$ to (x_0,t_0), and $(x_0+ct_0,0)$ to (x_0,t_0), respectively. We evaluate the integral seprately:

- On L_0, we have $t=0$, so $dt=0$. Also $u_t(x,t) = u_t(x,0) = \psi(x)$, so we get

$$\int_{L_0} (-c^2 u_x\ dt - u_t\ dx) = \int_{L_0} (-u_t dx) = -\int_{x-ct}^{x+ct} \psi(x)\ dx$$

- The line segment L_1 is represented by $x+ct = x_0+ct_0$. This implies that $dx+cdt=0$. Arranging we get $dt = -dx/c$, or $dx = -cdt$. So

$$\int_{L_1} (-c^2 u_x\ dt - u_t\ dx) = \int_{L_1} (cu_x\ dx + cu_t\ dt) = c\int_{L_1} (u_x\ dx + u_t\ dt)$$

  Recall by chain rule we see that $du = u_x dx + u_t dt$. So

$$\begin{aligned}\int_{L_1} (-c^2 u_x\ dt - u_t\ dx) &= c\int_{L_1} 1du \\ &= cu(x_0,t_0) - cu\,(x_0+ct_0,0) \\ &= cu(x_0,t_0) - c\phi(x_0+ct_0)\end{aligned}$$

- The line segment L_2 is represented by $x-ct = x_0-ct_0$. This implies that $dx-cdt=0$. Arranging we get $dt = dx/c$, or $dx = cdt$. So

$$\int_{L_2} (-c^2 u_x\ dt - u_t\ dx) = \int_{L_2} (-cu_x\ dx - cu_t\ dt) = -c\int_{L_2} (u_x\ dx + u_t\ dt)$$

  Recall by chain rule we see that $du = u_x dx + u_t dt$. So

$$\begin{aligned}\int_{L_2} (-c^2 u_x\ dt - u_t\ dx) &= c\int_{L_2} 1du \\ &= -cu(x_0-ct_0,0) + cu\,(x_0,t_0) \\ &= cu(x_0,t_0) - c\phi(x_0+ct_0)\end{aligned}$$

Therefore, adding up all terms together, we get

$$\iint_\triangle d\ dxdt = -\int_{x-ct}^{x+ct} \psi(x)\ dx + 2cu(x_0,t_0) - 2c\phi(x_0+ct_0)$$

We see u appears in second term of RHS. The required solution formula follows from isolating it out. Take note that (x_0,t_0) is to represent any point in the half plane $t \geq 0$, thus we can change them back to (x,t). This completes the proof. $\square$

Similarly, we can study the wave with a source on a half-line, which could be studied as Dirichlet problem or Neumann problem:

Definition 3.4.3 (Dirichlet wave with source on half-line)**.** The wave with a source on a half-line is modelled by the following system:

$$\begin{cases} v_{tt} - c^2 v_{xx} = f(x,t), \ x \in \mathbb{R}^+, \ t > 0 \\ v(0,t) = h(t) \\ v(x,0) = \phi(x) \\ v_t(x,0) = \psi(x) \end{cases}$$

Again we use the substitution method: let $V(x,t) := v(x,t) - h(t)$, then it satisfies

$$\begin{cases} V_{tt} - c^2 V_{xx} = f(x,t) - h''(t), \ x \in \mathbb{R}^+, \ t > 0 \\ V(0,t) = 0 \\ V(x,0) = \phi(x) - h(0) \\ V_t(x,0) = \psi(x) - h'(0) \end{cases}$$

This problem has a homogenous boundary condition, allowing the use of the method of reflection and odd extension. For simplicity let $F(x,t) := f(x,t) - h''(t)$, $\Phi(x) = \phi(x) - h(0)$, and $\Psi(x) = \psi(x) - h'(0)$, and so the system is now:

$$\begin{cases} V_{tt} - c^2 V_{xx} = F(x,t), \ x \in \mathbb{R}^+, \ t > 0 \\ V(0,t) = 0 \\ V(x,0) = \Phi(x) \\ V_t(x,0) = \Psi(x) \end{cases}$$

Let $\tilde{F}, \tilde{\Phi}, \tilde{\Psi}$ be the odd extension of F, Φ, Ψ, repsectively. We then solve the extended wave equation on the $\mathbb{R}$:

$$\begin{cases} U_{tt} - c^2 V_{xx} = \tilde{F}(x,t), \ x \in \mathbb{R}, \ t > 0 \\ U(0,t) = 0 \\ U(x,0) = \tilde{\Phi}(x) \\ U_t(x,0) = \tilde{\Psi}(x) \end{cases}$$

By the solution formula given in previous section, the solution is

$$U(x,t) = \frac{1}{2}[\tilde{\Phi}(x+ct) + \tilde{\Phi}(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \tilde{\Psi}(y)\ dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} \tilde{F}(y,s)\ dyds$$

The required solution $U(x,t) = V(x,t)$ is restricted on $x > 0$, which we now explicitly construct it. As usual, this has to be separate into two cases.

Firstly, when $x - ct > 0$, i.e. $x > ct$, then we get

$$V(x,t) = \frac{1}{2}[\Phi(x+ct) + \Phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \Psi(y)\ dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} F(y,s)\ dyds$$

Expanding out V, Φ, Ψ, and F, upon rearranging we get

$$v(x,t) = \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(y)\ dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(y,s)\ dyds$$

Secondly, when $x - ct < 0$, i.e. $x < ct$, then you get a notoriously tedious and complicated integration process which I don't think it's worth my effort typing out, thus I will only show you the final answer:

$$\begin{aligned} v(x,t) = h\left(t - \frac{x}{c}\right) &+ \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(y)\ dy \\ &+ \frac{1}{2c}\left\{\int_{t-\frac{x}{c}}^{t} \int_{x-c(t-s)}^{x+c(t-s)} + \int_0^{t-\frac{x}{c}} \int_{c(t-s)-x}^{x+c(t-s)}\right\} f(y,s)\ dyds \end{aligned}$$

Definition 3.4.4 (Neumann wave with source on half-line)**.** The wave with a source on a half-line is modelled by the following system:

$$\begin{cases} w_{tt} - c^2 w_{xx} = f(x,t), \ x \in \mathbb{R}^+, \ t > 0 \\ w_x(0,t) = g(t) \\ w(x,0) = \phi(x) \\ w_t(x,0) = \psi(x) \end{cases}$$

The Neumann problem of is similar, where substituting $W(x,t) = w(x,t) - xg(t)$ reduces the problem into homogenous boundary condition. Then use the reflection method to get the solution formula.

Chapter 4

Boundary Problems

Before starting the chapter, we give a short review on the following question, which will be used later.

Supposed we want to solve the following ODE:

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(\ell) = 0 \end{cases}$$

where $\lambda \in \mathbb{R}$ and $\ell > 0$. It is known that the characteristic equation (in terms of Λ) is

$$\Lambda^2 + \lambda = 0$$

1. If $\lambda < 0$, then $\Lambda_{1,2} = \pm\sqrt{-\lambda}$, which gives us solution

   $$X(x) = c_1 e^{\sqrt{-\lambda}x} + c_2 e^{-\sqrt{-\lambda}x}$$

   Substituting into the given initial conditions we get

   $$\begin{cases} c_1 + c_2 & = 0 \\ c_1 e^{\sqrt{-\lambda}\ell} + c_2 e^{-\sqrt{-\lambda}\ell} & = 0 \end{cases}$$

   Together this gives $c_1 = c_2 = 0$, so $X(x) = 0$.

2. If $\lambda = 0$, then $\Lambda_{1,2} = 0$, which gives us solution

   $$X(x) = c_1 e^{0x} + c_2 x e^{0x} = c_1 + c_2 x$$

   Solving out c_1 and c_2 using the given initial conditions give $c_1 = c_2 = 0$, so $X(x) = 0$.

3. If $\lambda > 0$, then $\Lambda_{1,2} = \pm\sqrt{-\lambda}i$, which gives us solution

   $$X(x) = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x)$$

   From $X(0) = 0$ we get $c_1 = 0$. Together with the condition $X(\ell) = 0$, we get

   $$c_2 \sin(\sqrt{\lambda}\ell) = 0 \implies c_2 = 0 \quad \text{or} \quad \sin\sqrt{(}\lambda\ell) = 0$$

   Note that $c_2 = 0$ leads to trivial solution. On the other hand:

   $$\begin{aligned} \sin\sqrt{(}\pi\ell) = 0 &\implies \sqrt{\lambda}\ell = k\pi, \quad k \in \mathbb{Z}^+ \\ &\implies \sqrt{\lambda} = \frac{k\pi}{\ell} \\ &\implies \sqrt{\lambda} = \frac{k^2\pi^2}{\ell^2} \end{aligned}$$

Thus the non-trivial solution is

$$X(x) = c_2 \sin \frac{k\pi x}{\ell}, \quad k \in \mathbb{Z}^+$$

In this case, to be specific, for every $k \in \mathbb{Z}^+$ we define $\lambda_k = \frac{k^2\pi^2}{\ell^2}$ and

$$X_k(x) = \sin \frac{k\pi x}{L}$$

Then

$$X(x) = \sum_{k=1}^{\infty} X_k(x) = \sum_{k=1}^{\infty} \sin \frac{k\pi x}{\ell}$$

is also a solution if and only if the RHS converges.

This finishes the preparation.

4.1 Boundary Value Problems

We have been studying PDEs on whole line or half line. However, realistically many physical problems have boundaries. This chapter studies the boundary value problems of PDEs.

We start from wave equation:

Definition 4.1.1 (Homogeneous wave equation with Dirichlet boundary conditions)**.** The wave equation $u_{tt} = c^2 u_{xx}$ where $0 < x < \ell$ and $t > 0$ is subjected to the homogeneous boundary conditions

$$u(0,t) = u(\ell,t) = 0, \quad t > 0$$

and the initial condition

$$u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad 0 < x < \ell$$

where $c, \ell > 0$ are given constants, and ϕ, ψ are given functions.

The whole setting is analogous to all of our previous study, except that now there is a restriction on the value of x, setting two 'walls' at $x = 0$ and $x = \ell$.

The strategy of solving such a problem is called *separaction of variables*, where we first look for a separated solution in the form of $u(x,t) = X(x)T(t)$, which substituting into the wave equation will reduce into ODEs for X and T.

Theorem 4.1.2. *The homogeneous wave equation with Dirichlet boundary conditions has the following series solution:*

$$u(x,t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi ct}{\ell} + B_n \sin \frac{n\pi ct}{\ell} \right) \sin \frac{n\pi x}{\ell}$$

where A_n and B_n are to be determined by the initial condition.

Proof. Assuming we have the solution of the form $u(x,t) = X(x)T(t)$, substituting back to the equation leads to

$$X(x)T''(t) = c^2 X''(x)T(t) \implies -\frac{X''(x)}{X(x)} = -\frac{T''(t)}{c^2 T(t)} := \lambda \tag{4.1}$$

where the minus sign is inserted for later convenience. We claim that λ is a constant, which can be seen by noting that x and t are independent variables, so

$$\frac{\partial \lambda}{\partial t} = \frac{\partial}{\partial t}\left(-\frac{X''(x)}{X(x)}\right) = 0 \quad \text{and} \quad \frac{\partial \lambda}{\partial x} = \frac{\partial}{\partial x}\left(-\frac{T''(x)}{c^2 T(t)}\right) = 0$$

We also prove that $\lambda > 0$:

1. If $\lambda = 0$, Equation (4.1) gives $X'' = 0$, so $X(x) = ax + b$. The boundary condition $u(0,t) = u(\ell,t) = 0$ implies

   $$X(0)T(t) = X(\ell)T(t) = 0$$

   and so $X(0) = X(\ell) = 0$. Solving out we get $a = b = 0$, giving $X(x) = 0$, thus leads to trivial solution, which we are not interested at.

2. If $\lambda < 0$, Equation (4.1) gives
$$X''(x) + \lambda X(x) = 0$$
where by the first case of our previous preparation we know we have trivial solution, which we are also not interested at.

With this we can now safely assume that $\lambda = \beta^2$ for some $\beta > 0$. We thsu obtain a pair of separate ODEs from Equation (4.1):
$$\begin{cases} X''(x) + \beta^2 X(x) = 0 \\ T''(t) + c^2\beta^2 T(t) = 0 \end{cases}$$
The first ODE in terms of x has general solution
$$X(x) = C\cos\beta x + D\sin\beta x$$
where C and D are arbitrary constant. Recall that the boundary conditions demand $u(0,t) = u(\ell,t) = 0$, i.e. $X(0)T(t) = X(\ell)T(t) = 0$. If $T(t) = 0$ it clear that $u(x,t) = X(x)T(t) = 0$ is the trivial solution. So $T(t) \neq 0$, and we have $X(0) = X(\ell) = 0$, i.e.
$$\begin{cases} C & = 0 \\ D\sin\beta\ell & = 0 \end{cases}$$
If $D = 0$, we ended up with trivial solution, so we consider $\sin\beta\ell - 0$. This implies
$$\beta\ell = n\pi,\ n \in \mathbb{Z} \implies \beta = \frac{n\pi}{\ell},\ n \in \mathbb{Z}$$
Recall that $\lambda = \beta^2$, so similar to what we did in the previous preparation, let
$$\lambda_n = \left(\frac{n\pi}{\ell}\right)^2 \quad \text{and} \quad X_n(x) = \sin\frac{n\pi x}{\ell}$$
We see that for each $n \in \mathbb{Z}$ the pair (λ_n, X_n) is a solution to the problem
$$X'' = -\lambda X, \quad X(0) = X(\ell) = 0$$
We now turn our attention to the second ODE $T'' + c^2\beta^2 T = 0$. With the same argument, we can determind the form of its general solution and find that the roots of the characteristic equation are
$$\pm\frac{n\pi c}{\ell}i$$
and thus
$$T_n(t) = A_n\cos\frac{n\pi ct}{\ell} + B_n\sin\frac{n\pi ct}{\ell}$$
for each $n \in Z^+$, where A_n and B_n are arbitrary constant.

Thus, the wave equation with boundary conditions $u(0,t) = u(\ell,t) = 0$ has solution, in fact, infinite number of separated solutions:
$$u(x,t) = X_n(x)T_n(t) = \sin\frac{n\pi x}{\ell}\left(A_n\cos\frac{n\pi ct}{\ell} + B_n\sin\frac{n\pi ct}{\ell}\right)$$
for each $n \in \mathbb{Z}^+$. Based on the principle of superposition, the series
$$u(x,t) = \sum_{n=1}^{\infty}\sin\frac{n\pi x}{\ell}\left(A_n\cos\frac{n\pi ct}{\ell} + B_n\sin\frac{n\pi ct}{\ell}\right)$$
is also a solution.

Lastly, we have to look for A_n and B_n using the initial condition $u(x,0) = \phi(x)$ and $u_t(x,0) = \psi(x)$. These two conditions give two equations:
$$\phi(x) = \sum_{n=1}^{\infty} A_n\sin\frac{n\pi x}{\ell}$$

$$\psi(x) = \sum_{n=1}^{\infty} \frac{n\pi c}{\ell} B_n \sin \frac{n\pi x}{\ell}$$

The determination of A_n and B_n requires the technique of Fourier series, which will be covered in Chapter 5. The proof is considered completed at the moment. □

Remark 4.1.3 (Coefficients of t and frequency)**.** The coefficient of t inside the series solution is $n\pi c/\ell$. These are called the *frequencies*. The smallest frequecy $\pi c/\ell$ is called the fundamental note. All integer multiples of the fundamental notes, such as $2\pi c/\ell$ and $3\pi c/\ell$, are called the overtones. Notice that shortening the length of ℓ results in larger frequency. This matched with the physical phenomenon where plucking a shorter length of string in any string instrument gives a sharper (higher frequency) sound.

Of course, we need not restrict ourselves to study merely the wave equation. The following is an analogous formulation of problem in terms of heat equation

Definition 4.1.4 (Homogeneous heat equation with Dirichlet boundary conditions)**.** The heat equation $u_t = ku_{xx}$ where $0 < x < \ell$ and $t > 0$ is subjected to the homogeneous boundary conditions

$$u(0,t) = u(\ell,t) = 0, \quad t > 0$$

and the initial condition

$$u(x,0) = \phi(x), \quad 0 < x < \ell$$

where $k, \ell > 0$ are positive constants, and ϕ is given function.

It is natural to interpret it as having our usual heat conduction, except that the body now has finite length, and the two ends of the body is kept cold. The same solution technique above is applied to obtain the solution in this case:

Theorem 4.1.5. *The homogeneous heat equation with Dirichlet boundary conditions has solution*

$$u(x,t) = \sum_{n=1}^{\infty} A_n \exp\left(-\left(\frac{n\pi}{\ell}\right)^2 kt\right) \sin \frac{n\pi x}{\ell}$$

provided that ϕ has the expansion

$$\phi(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{\ell}$$

Proof sketch. We only provide a sketch here, since the overall idea is similar to the case of wave equation. First, suppose that the solution takes the form $u(x,t) = X(x)T(t)$, which we get

$$-\frac{X''}{X} = -\frac{T'}{kT} := \lambda$$

Again λ is a constant. We have the eigenproblem

$$X'' = -\lambda X$$

with condition $X(0) = X(\ell) = 0$. There are infinitely many solutions pair:

$$\lambda_n = \left(\frac{n\pi}{\ell}\right)^2 \quad \text{and} \quad X_n(x) = \sin \frac{n\pi x}{\ell}$$

for each $n \in \mathbb{Z}^+$. On the other hand, the ODE

$$T' + \lambda kT = 0$$

has solutions, for each $n \in \mathbb{Z}^+$, that

$$T_n(t) = A_n \exp\left(-\left(\frac{n\pi}{\ell}\right)^2 kt\right)$$

The solution follows directly. Of course, we have to solve out A_n based on the provided initial condition $u(x,0) = \phi(x)$, and this will be covered in the later chapter. □

4.2 Separation of Variables: The Neumann Condition

In this section we use the method of separation to solve wave equation and heat equation under the Neumann condition.

Theorem 4.2.1. *The homogeneous wave equation with Neumann boundary conditions:*

$$u_{tt} = c^2 u_{xx},\ 0 < x < \ell,\ t > 0,$$

$$u_x(0,t) = u_x(\ell,t) = 0,\ t > 0,$$

$$u(x,0) = \phi(x),\ u_t(x,0) = \psi(x),\ 0 < x < \ell$$

has the following series solution:

$$u(x,t) = \frac{A_0 + B_0 t}{2} + \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi ct}{\ell} + B_n \sin \frac{n\pi ct}{\ell} \right) \cos \frac{n\pi x}{\ell}$$

provided that the initial datas can be expanded into cosine Fourier series

$$\phi(x) = \frac{1}{2} A_0 + \sum_n A_n \cos \frac{n\pi x}{\ell}$$

$$\psi(x) = \frac{1}{2} B_0 + \sum_n \frac{n\pi c}{\ell} B_n \cos \frac{n\pi x}{\ell}$$

Proof sketch. The procedure is the same, where subtituting $u(x,t) = X(x)T(t)$ obtains

$$X'' = -\lambda X \quad \text{and} \quad -T'' = c^2 \lambda T$$

The first difference is that the boundary condition imply

$$X'(0)T(t) = X'(\ell)T(t) = 0$$

Since $T(t) = 0$ results in trivial solution, we see that we should have

$$X'(0) = X'(\ell) = 0$$

Again, we focus on the case $\lambda \geq 0$. Let $\lambda = \beta^2$, the general solution is

$$X(x) = C \cos \beta x + D \sin \beta x$$

Differentiating $X(x)$ and using the condittion $X'(0) = X'(\ell) = 0$, we get $D = 0$ and $C \sin \beta\ell = 0$. We don't want $C = 0$ as it gives trivial solution, so consider $\sin \beta\ell = 0$, which implies $\beta\ell = n\pi$ for any $n \in \mathbb{Z}$. This gives solution pair

$$\lambda_n = \left(\frac{n\pi}{\ell} \right)^2 \quad \text{and} \quad X_n(x) = \cos \frac{n\pi x}{\ell}$$

where $n \in \mathbb{Z}_0^+$. For $T(t)$, solving $T'' = -c^2 \lambda T$ gives the formula

$$T_n(t) = A_n \cos \frac{n\pi ct}{\ell} + B_n \sin \frac{n\pi ct}{\ell}$$

for $n \geq 1$, while

$$T_0(t) = \frac{1}{2} A_0 + \frac{1}{2} B_0 t$$

□

For heat equation, we simply give the solution formula. The proof is omitted.

Theorem 4.2.2. *The homogeneous heat equation with Neumann boundary conditions:*

$$u_t = ku_{xx},\ 0 < x < \ell,\ t > 0,$$

$$u_x(0,t) = u_x(\ell,t) = 0,\ t > 0,$$

$$u(x,0) = \phi(x),\ 0 < x < \ell$$

has the following series solution:

$$u(x,t) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n \exp\left(-\left(\frac{n\pi}{\ell}\right)^2 kt\right)\cos\frac{n\pi x}{\ell}$$

provided that ϕ can be expanded into cosine Fourier series

$$\phi(x) = \frac{1}{2}A_0 + \sum_n A_n \cos\frac{n\pi x}{\ell}$$

4.3 Mixed Boundary Conditions

We give a taste of mixed boundary condition problem, before treating them in a more serious manner in Chapter 5. An example of mixed boundary conditions is having Dirichlet conditions at one end of the finite interval, and Neumann conditions at the other.

For example, consider the heat equation with mixed boundary conditions:

$$u_t = ku_{xx},\ 0 < x < \ell,\ t > 0,$$

$$u(0,t) = u_x(\ell,t) = 0,\ t > 0,$$

$$u(x,0) = \phi(x),\ 0 < x < \ell$$

To get the solution formula, we again assume that $u(x,t) = X(x)T(t)$ and by substitution we get

$$X'' + \lambda X = 0$$

where the provided initial boundary conditions implies that $X(0) = X'(\ell) = 0$.

For $\lambda > 0$, the solution structure is

$$X(x) = C\cos(\sqrt{\lambda}x) + D\sin(\sqrt{\lambda}x)$$

The condition $X(0) = 0$ implies $C = 0$, while the condition $X'(\ell) = 0$ says that $D = 0$ or $\cos(\sqrt{\lambda}\ell) = 0$. We do not want the trivial solution, so we consider $\cos(\sqrt{\lambda}\ell) = 0$. In general, the solutions are

$$\lambda_n = \left(n + \frac{1}{2}\right)^2 \frac{\pi^2}{\ell^2} \quad \text{and} \quad X_n(x) = \sin\left(\left(n + \frac{1}{2}\right)\frac{\pi x}{\ell}\right)$$

where $n \in \mathbb{Z}_0^+$.

For $\lambda = 0$, we se that $X'' = 0$ and $X(x) = ax + b$, and thus $X(0) = b = 0$, and $X'(\ell) = a = 0$. This gives trivial solution, which we also reject.

For $\lambda < 0$, the solution of the ODE is

$$X(x) = Ce^{-\sqrt{-\lambda}x} + De^{\sqrt{-\lambda}x}$$

Again, using the boundary conditions we can show that $C = D = 0$, thus rejected as well.

On the other hand, the equation for $T(t)$ is still the same as all our previous case. Therefore we get the following solution formula:

$$u(x,t) = \sum_{n=0}^{\infty} A_n \exp\left(-\left(n + \frac{1}{2}\right)^2 \frac{\pi^2 kt}{\ell^2}\right)\sin\left(\left(n + \frac{1}{2}\right)\frac{\pi x}{\ell}\right)$$

where A_n is determined by ϕ.

Chapter 5

Fourier Series

In the previous chapter, we encounter problems of expanding coefficients in the derived solution formula. Here, we will see that this problem will be solved by the technique of Fourier Series. Again, just like previous chapter, we do some setup work that will ease our later computations.

We first compute

$$I_1 = \int_0^L \sin\frac{n\pi x}{L}\sin\frac{m\pi x}{L}\ dx \text{ where } n, m \in \mathbb{Z}^+$$

Note that by basic trigo identity we can simplify the integrand into

$$\begin{aligned}\sin\frac{n\pi x}{L}\sin\frac{m\pi x}{L} &= \frac{1}{2}\cos\left(\frac{n\pi x}{L} - \frac{m\pi x}{L}\right) - \frac{1}{2}\cos\left(\frac{n\pi x}{L} + \frac{m\pi x}{L}\right)\\ &= \frac{1}{2}\cos\frac{n-m}{L}\pi x - \frac{1}{2}\cos\frac{n+m}{L}\pi x\end{aligned}$$

and so the integral become

$$I_1 = \frac{1}{2}\int_0^L \cos\frac{n-m}{L}\pi x\ dx - \frac{1}{2}\int_0^L \cos\frac{n+m}{L}\pi x\ dx$$

We split into two cases:

- If $n = m$, then simply

$$I_1 = \frac{1}{2}\int_0^L 1\ dx - \frac{L}{2n\pi}\sin\frac{2n\pi x}{L}\bigg|_0^L = \frac{1}{2}L - 0 = \frac{1}{2}L$$

- If $n \neq m$, then we have

$$I_1 = \frac{1}{2}\frac{L}{(n-m)\pi}\sin\frac{n-m}{L}\pi x\bigg|_0^L - \frac{1}{2}\frac{L}{(n+m)\pi}\sin\frac{n+m}{L}\pi x\bigg|_0^L = 0$$

In short we have

$$I_1 = \int_0^L \sin\frac{n\pi x}{L}\sin\frac{m\pi x}{L}\ dx = \begin{cases} 0 & ,\ n \neq m \\ \frac{L}{2} & , n = m \end{cases}$$

Next, we compute

$$I_2 = \int_0^L \cos\frac{n\pi x}{L}\cos\frac{m\pi x}{L}\ dx \text{ where } n, m \in \mathbb{Z}^+$$

Similarly, by arranging the integrand we have

$$I_2 = \frac{1}{2}\int_0^L \cos\frac{n+m}{L}\pi x\ dx + \frac{1}{2}\int_0^L \cos\frac{n+m}{L}\pi x\ dx$$

Splitting into cases:

- When $n = m = 0$, we have $I = L$.
- When $n \neq m$, then

$$I_2 = \frac{1}{2}\frac{L}{(n+m)\pi}\sin\frac{(n+m)\pi}{L}x\bigg|_0^L + \frac{1}{2}\frac{L}{(n-m)\pi}\sin\frac{(n-m)\pi}{L}x\bigg|_0^L = 0$$

- When $n = m \neq 0$, then

$$\begin{aligned} I_2 &= \frac{1}{2}\int_0^L \cos\frac{2n\pi}{L}x\ dx + \frac{1}{2}\int_0^L 1\ dx \\ &= \frac{1}{2}\frac{L}{2n\pi}\sin\frac{2n\pi}{L}x\bigg|_0^L + \frac{1}{2}L \\ &= \frac{1}{2}L \end{aligned}$$

In short, we have

$$I_2 = \int_0^L \cos\frac{n\pi x}{L}\cos\frac{m\pi x}{L}\ dx = \begin{cases} L & ,\ n = m = 0 \\ \frac{L}{2} & ,n = m > 0 \\ 0 & ,n \neq m \end{cases}$$

We compute

$$I_3 = \int_{-L}^L \cos\frac{n\pi x}{L}\sin\frac{m\pi x}{L}\ dx \text{ where } n \in \mathbb{Z}_0^+,\ m \in \mathbb{Z}^+$$

We shall omit the calculation process. The final result is that

$$I_3 = \int_{-L}^L \cos\frac{n\pi x}{L}\sin\frac{m\pi x}{L}\ dx = 0$$

Lastly, we compute that

$$\begin{aligned} I_4 = \int_{-L}^L \cos\frac{n\pi x}{\ell}\cos\frac{m\pi x}{\ell}\ dx &= \int_0^L \cos\frac{n\pi x}{\ell}\cos\frac{m\pi x}{\ell}\ dx - \int_0^{-L} \cos\frac{n\pi x}{\ell}\cos\frac{m\pi x}{\ell}\ dx \\ &= \int_0^L \cos\frac{n\pi x}{\ell}\cos\frac{m\pi x}{\ell}\ dx - \int_0^L \cos\frac{n\pi x}{\ell}\cos\frac{m\pi x}{\ell}\ dx \\ &= 0 \end{aligned}$$

and

$$\begin{aligned} I_5 = \int_{-L}^L \sin\frac{n\pi x}{\ell}\sin\frac{m\pi x}{\ell}\ dx &= \int_0^L \sin\frac{n\pi x}{\ell}\sin\frac{m\pi x}{\ell}\ dx - \int_0^{-L} \sin\frac{n\pi x}{\ell}\sin\frac{m\pi x}{\ell}\ dx \\ &= \int_0^L \sin\frac{n\pi x}{\ell}\sin\frac{m\pi x}{\ell}\ dx - \int_0^L \sin\frac{n\pi x}{\ell}\sin\frac{m\pi x}{\ell}\ dx \\ &= 0 \end{aligned}$$

5.1 Fourier Sine Series

Recall that a function f is *periodic* if

$$\exists p \in \mathbb{Z}^+ \text{ such that } f(x+p) = f(x)\ \forall x \in \mathrm{Dom}_f$$

Here p is called the period of $f(x)$. A good reason for mentioning this property is because both sine and cosine functions are periodic. In particular, both $\sin\omega x$ and $\cos\omega x$ are $\frac{2\pi}{\omega}$-periodic. The periodic property of sine and cosine function can help in simplfying the computations.

We have encountered the Fourier sine series in the previous chapter. We give a formal definition here:

Definition 5.1.1 (Fourier Sine Series)**.** Suppose $\phi(x)$ is a given function with domain $(0, \ell)$. The Fourier sine series of ϕ over $(0, \ell)$ is defined to be

$$\phi(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{\ell}$$

where $0 < x < \ell$ and A_n are some coefficients.

The goal here is to find the coefficients A_n. To do so, recall that the eigeproblem

$$X''(x) = -\lambda X(x) \quad \text{and} \quad X(0) = X(\ell) = 0$$

has eigenfunctions, i.e. solutions, as

$$\left\{ \sin \frac{\pi x}{\ell},\ \sin \frac{2\pi x}{\ell}, \ldots, \sin \frac{n\pi x}{\ell}, \ldots \right\}$$

Recall that we have the following computation result:

$$I_1 = \int_0^L \sin \frac{n\pi x}{L} \sin \frac{m\pi x}{L}\ dx = \begin{cases} 0 & ,\ n \neq m \\ \frac{L}{2} & ,n = m \end{cases}$$

This means that the series of sine functions are mutually orthogonal. Another clear observation says that they are linearly independent, as no one term is a multiple of another.

We now determine the coefficients A_n. Suppose given the Fourier sine series of expansion of $\phi(x)$:

$$\phi(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{\ell}$$

where $0 < x < \ell$. Multiply both sides by $\sin \frac{m\pi x}{\ell}$ and take integration over $(0, \ell)$ we get

$$\int_0^\ell \phi(x) \sin \frac{m\pi x}{\ell}\ dx = \sum_{n=1}^{\infty} A_n \int_0^\ell \sin \frac{n\pi x}{\ell} \sin \frac{m\pi x}{\ell}\ dx$$

By orthogonality, we can evaluate the RHS of the equation:

$$\int_0^\ell \phi(x) \sin \frac{m\pi x}{\ell}\ dx = A_m \frac{\ell}{2}$$

Arranging, we get

$$A_m = \frac{2}{\ell} \int_0^\ell \phi(x) \sin \frac{m\pi x}{\ell}\ dx$$

Since m is arbitrarily chosen, thus the above formula is applicable for all A_m with $m \in \mathbb{Z}^+$. With this, we can now provide the full solution formula:

Theorem 5.1.2. *The homogeneous heat equation with Dirichlet boundary conditions*

$$u_t = k u_{xx},\ 0 < x < \ell,\ t > 0$$

$$u(0, t) = u(\ell, t) = 0,\ t > 0$$

$$u(x, 0) = \phi(x),\ 0 < x < \ell$$

where $k, \ell \in \mathbb{R}^+$ and ϕ is given function, has solution formula

$$u(x, t) = \sum_{n=1}^{\infty} A_n \exp\left(-\left(\frac{n\pi}{\ell}\right)^2 kt \right) \sin \frac{n\pi x}{\ell}$$

where the coefficients A_n are given by

$$A_n = \frac{2}{\ell} \int_0^\ell \phi(x) \sin \frac{n\pi x}{\ell}\ dx$$

provided that the series formula $u(x, t)$ converges

Remark 5.1.3 (Validity of the solution formula)**.** To emphasize again, despite we have determined the coefficients A_n, the solution formula $u(x,t)$ is only valid given that the series converges.

Despite our goal is to expand the coefficients in the solution formula, expanding a given function into Fourier series is often useful in many fields. We use an example to demonstrate the process.

Example 5.1.4. We expand $\phi(x) = 1$ where $x = (0, \ell)$ in Fourier sine series. By the derivation, the coefficient is

$$A_n = \frac{2}{\ell}\int_0^\ell 1\ \sin\frac{n\pi x}{\ell}\ dx = -\frac{2}{n\pi}\cos\frac{n\pi x}{\ell}\Big|_0^\ell = \frac{2}{n\pi}[1-(-1)^n]$$

where $n \in \mathbb{Z}^+$. When n is even, we get $A_n = 0$. When n is odd, we get $A_n = \frac{4}{n\pi}$. Thus the Fourier series expansion in $(0, \ell)$ is given by

$$1 = \sum_{n=1}^{\infty} A_n \sin\frac{n\pi x}{\ell} = \frac{4}{\pi}\left(\sin\frac{\pi x}{\ell} + \frac{1}{3}\sin\frac{3\pi x}{\ell} + \frac{1}{5}\sin\frac{5\pi x}{\ell} + \dots\right)$$

Example 5.1.5. We expand $\phi(x) = x$ where $x \in (0, \ell)$ in Fourier sine series. Evaluate the coefficents using integration by parts we get:

$$\begin{aligned} A_n &= \frac{2}{\ell}\int_0^\ell x\sin\frac{n\pi x}{\ell}\ dx \\ &= -\frac{2}{n\pi}\cos\frac{n\pi x}{\ell}\Big|_0^\ell + \frac{2}{n\pi}\int_0^\ell \cos\frac{n\pi x}{\ell}\ dx \\ &= \frac{2\ell}{n\pi}(-1)^{n+1} \end{aligned}$$

Thus the Fourier sine expansion in $(0, \ell)$ is given by

$$x = \frac{2\ell}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin\frac{n\pi x}{\ell} = \frac{2\ell}{\pi}\left(\sin\frac{\pi x}{\ell} - \frac{1}{2}\sin\frac{2\pi x}{\ell} + \frac{1}{3}\sin\frac{3\pi x}{\ell}\right)$$

5.2 Fourier Cosine Series

We give a formal definition for Fourier cosine series:

Definition 5.2.1 (Fourier Cosine Series)**.** Suppose $\phi(x)$ is a given function with domain $(0, \ell)$. The Fourier cosine series of ϕ over $(0, \ell)$ is defined to be

$$\phi(x) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n \cos\frac{n\pi x}{\ell}$$

where $0 < x < \ell$ and A_n are some coefficients.

The properties of the system

$$\left\{1,\ \cos\frac{\pi x}{\ell},\ \cos\frac{2\pi x}{\ell} + \dots + \cos\frac{n\pi x}{\ell}, \dots\right\}$$

is muchly the same as the previously discussed system of sine functions. The orthogonality here is given by

$$I_2 = \int_0^L \cos\frac{n\pi x}{L}\cos\frac{m\pi x}{L}\ dx = \begin{cases} L & ,\ n = m = 0 \\ \frac{L}{2} & , n = m > 0 \\ 0 & , n \neq m \end{cases}$$

We now determine the coefficients A_n in the cosine series. Multiplying $\cos\frac{m\pi x}{\ell}$ on both sides of Fourier cosine series of ϕ, then take integration over $(0, \ell)$, we obtain

$$\int_0^\ell \phi(x)\cos\frac{m\pi x}{\ell}\ dx = \frac{1}{2}A_0\int_0^\ell \cos\frac{m\pi x}{\ell}\ dx + \sum_{n=1}^{\infty} A_n\int_0^\ell \cos\frac{n\pi x}{\ell}\cos\frac{m\pi x}{\ell}\ dx$$

By orthogonality and some algebraic manipulations, it implies that

$$A_m = \frac{2}{\ell}\int_0^\ell \phi(x)\cos\frac{m\pi x}{\ell}\ dx$$

Again, since A_m is arbitrary, the coefficients formula applies for all $m \in \mathbb{Z}_0^+$. We now get the full solution formula:

Theorem 5.2.2. *The homogeneous heat equation with Neumann boundary conditions*

$$u_t = ku_{xx},\ 0 < x < \ell,\ t > 0$$

$$u_x(0,t) = u_x(\ell,t) = 0,\ t > 0$$

$$u(x,0) = \phi(x),\ 0 < x < \ell$$

where $k, \ell \in \mathbb{R}^+$ and ϕ is given function, has solution formula

$$u(x,t) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n \exp\left(-\left(\frac{n\pi}{\ell}\right)^2 kt\right)\cos\frac{n\pi x}{\ell}$$

where the coefficients A_n are given by

$$A_n = \frac{2}{\ell}\int_0^\ell \phi(x)\cos\frac{m\pi x}{\ell}\ dx$$

provided that the series formula $u(x,t)$ converges

The same remark is also applicable here: the solution formula is valid only if the series converges.

Next, we provide some examples of expanding a given function into Fourier cosine series:

Example 5.2.3. We expand $\phi(x) = 1$ where $x = (0,\ell)$ in Fourier cosine series. By the derivation, the coefficient is

$$A_n = \frac{2}{\ell}\int_0^\ell \cos\frac{n\pi x}{\ell}\ dx$$

When $n = 0$, we have

$$A_0 = \frac{2}{\ell}\int_0^\ell 1\ dx = 2$$

When $n > 0$, we have

$$A_n = \frac{2}{n\pi}\sin\frac{n\pi x}{\ell}\Big|_0^\ell = \frac{2}{n\pi}[\sin n\pi - \sin 0] = 0$$

Thus the Fourier cosine expansion is trivial, in the sense that, by the formula:

$$1 = 1 + 0\cos\frac{\pi x}{\ell} + 0\cos\frac{2\pi x}{\ell} + \dots$$

Example 5.2.4. We expand $\phi(x) = x$ where $x \in (0,\ell)$ in Fourier cosine series. Evaluate the coefficents for $n = 0$ we get

$$A_0 = \frac{2}{\ell}\int_0^\ell x\ dx = \ell$$

For $n > 0$, we shall apply integration by parts to obtain:

$$\begin{aligned}
A_n &= \frac{2}{\ell}\int_0^\ell x\cos\frac{n\pi x}{\ell}\ dx \\
&= \frac{2}{n\pi}x\sin\frac{n\pi x}{\ell}\Big|_0^\ell - \frac{2}{n\pi}\int_0^\ell \sin\frac{n\pi x}{\ell}\ dx \\
&= 0 + \frac{2\ell}{n^2\pi^2}\cos\cos\frac{n\pi x}{\ell}\Big|_0^\ell \\
&= \frac{2\ell}{n^2\pi^2}[(-1)^n - 1]
\end{aligned}$$

See that when $n > 0$ is even we get $A_n = 0$, and when n is odd we get $A_n = -\frac{4\ell}{n^2\pi^2}$ Thus the Fourier cosine series expansion is

$$x = \frac{\ell}{2} - \frac{4\ell}{\pi^2}\sum_{k=0}^{\infty}\frac{1}{((2k+1)^2)}\cos\frac{(2k+1)\pi x}{\ell}$$

where we have eliminate all the zero terms in the initial sums, and rewrite the non-zero sums so that the summation starts from 0 and increases by 1.

5.3 Full Fourier Series

The full Fourier series is simply the combination of sine and cosine series. The formal definition is as follow:

Definition 5.3.1. (Fourier Series) The full Fourier series (or more commonly, Fourier series) of $\phi(x)$ on $(-\ell, \ell)$ is given as

$$\phi(x) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty}\left(A_n\cos\frac{n\pi x}{\ell} + B_n\sin\frac{n\pi x}{\ell}\right)$$

where the coefficients are given by:

$$A_n = \frac{1}{\ell}\int_{-\ell}^{\ell}\phi(x)\cos\frac{n\pi x}{\ell}\,dx, \quad n \in \mathbb{Z}_0^+$$

$$B_n = \frac{1}{\ell}\int_{-\ell}^{\ell}\phi(x)\sin\frac{n\pi x}{\ell}\,dx, \quad n \in \mathbb{Z}^+$$

The corresponding system to study here is the blend of the two previous:

$$\left\{1, \cos\frac{\pi x}{\ell}, \sin\frac{\pi x}{\ell}, \cos\frac{2\pi x}{\ell}, \sin\frac{2\pi x}{\ell}, \ldots, \cos\frac{n\pi x}{\ell}, \sin\frac{n\pi x}{\ell}, \ldots\right\}$$

Example 5.3.2. We expand $\phi(x) = x$ where $x \in [-\ell, \ell]$ in full Fourier series. The coefficients to solve are

$$A_n = \frac{1}{\ell}\int_{-\ell}^{\ell}x\cos\frac{n\pi x}{\ell}\,dx, \quad n \in \mathbb{Z}_0^+$$

and

$$B_n = \frac{1}{\ell}\int_{-\ell}^{\ell}x\sin\frac{n\pi x}{\ell}\,dx, \quad n \in \mathbb{Z}^+$$

Note that in A_n, the integrand $x\cos\frac{n\pi x}{\ell}$ is an odd function in the interval $(-\ell, \ell)$, so $A_n = 0$. On the other hand, for each B_n, performing integration by parts we get

$$B_n = \frac{1}{\ell}\int_{-\ell}^{\ell}x\sin\frac{n\pi x}{\ell}\,dx = \left(\frac{-x}{n\pi}\cos\frac{n\pi x}{\ell} + \frac{1}{n^2\pi^2}\sin\frac{n\pi x}{\ell}\right)\Bigg|_{-\ell}^{\ell} = (-1)^{n+1}\frac{2\ell}{n\pi}$$

and thus the full Fourier series of $\phi(x)$ is

$$x = \sum_{n=1}^{\infty}(-1)^{n+1}\frac{2\ell}{n\pi}\sin\frac{n\pi x}{\ell}$$

5.4 Completeness

Despite successfully deriving out the full solution formulas, we see that they are valid only when we have the series is converging. Thus, we study the convergence of the Fourier series.

We give the three different types of divergences. The notation S_N is used to denote the first N-th sum of the series.

Definition 5.4.1 (Pointwise Convergence)**.** The series $\sum f_n(x)$ converges pointwisely to $f(x)$ if each $x \in (a, b)$ we have

$$\lim_{N\to\infty} S_N(x) = f(x)$$

Definition 5.4.2 (Uniform Convergence)**.** The series $\sum f_n(x)$ converges uniformly to $f(x)$ if we have

$$\lim_{N\to\infty} \max_{x\in[a,b]} |f(x) - S_N(x)| = 0$$

Definition 5.4.3 (L^2-Convergence)**.** The series $\sum f_n(x)$ converges in the L^2-sense to $f(x)$ if we have

$$\lim_{N\to\infty} \int_a^b |f(x) - S_N(x)|^2 \, dx = 0$$